Ford Blue Advantage® Gold Certification

Blue Advantage

Updated as of January 2024. Any updates communicated via dealer communication on FMCDealer after this date take precedent.

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Specifications and descriptions contained within are based upon the most current information available at the time of publication.

Section 1

Inspection and Certification Process

Inspection and Certification Process

Certification of Ford vehicles requires a thorough inspection, as well as mechanical and appearance reconditioning:

- Eligible Ford Blue Advantage® Certified vehicles must pass a manufacturer's inspection of up to 172 points for Gold Certification as found in this guide. **Note:** Not all 172 points will be applicable to every vehicle.
- Request a Vehicle History Report using the program website to verify the eligibility of all vehicles to be certified; a copy of the CARFAX Vehicle History Report will be provided during the vehicle certification process and must be provided to the customer at delivery and a copy retained in the deal jacket for that vehicle
- Inspections are to be completed by an eligible technician (see Training Requirements section of this guide for additional details regarding Service Technician eligibility requirements)
  - All work for items noted by the technician on the Vehicle Inspection Checklist as needing "Repair" or "Replacement" must be completed prior to registering the vehicle as certified
    - Reference copies of the appropriately branded Vehicle Inspection Checklists are included in the Appendix of this guide; they can also be printed from the program website See Program Forms portion of the "Website Drill-Downs: Administration and Performance Tracking" section of this guide. Printed NCR versions of the appropriately branded Vehicle Inspection Checklists may also be ordered from the Dealer eStore. See the Dealer eStore portion of the "Website Drill-Downs: Administration and Performance Tracking" section of this guide for details
  - Genuine Ford/Motorcraft®/Omnicraft® new or Ford Authorized Remanufactured Parts, when available the next business day, must be used exclusively in the mechanical reconditioning process
  - Interior and exterior appearance reconditioning must meet standards outlined in the appropriate Vehicle Inspection Checklist
  - Any open recall(s) that could not be completed due to repair procedures or parts availability must be noted with the FSA/Recall number and description by the technician on the Ford Blue Advantage Gold 172-Point Vehicle Inspection Checklist as appropriate
- Additionally, the following operations must be completed for all Ford Blue Advantage Certified vehicles:
  - Motorcraft/Omnicraft oil/filter change*
  - Installation of new Motorcraft/Omnicraft windshield wiper blades,* when available the next business day
  - Fuel tank filled at time of delivery*
  - Replace tires if:
    - Tires are mismatched (Ford Blue Advantage Gold ONLY)
    - Tires have uneven wear
    - Remaining tread depth is less than 5/32 inch
- Brake pads/linings must have at least 50% thickness remaining
- Any parts installed on the vehicle must be Motorcraft/Omnicraft parts* when available the next business day

Important!

- Upon completion, the inspecting Service Technician must complete and sign the Vehicle Inspection Checklist and forward it to the Service Manager and Ford Blue Advantage Pre-Owned Vehicle Manager for their respective signatures. The signed and date-stamped original of the completed Vehicle Inspection Checklist must be provided to the owner at delivery. The dealer copy should be retained in the vehicle deal jacket

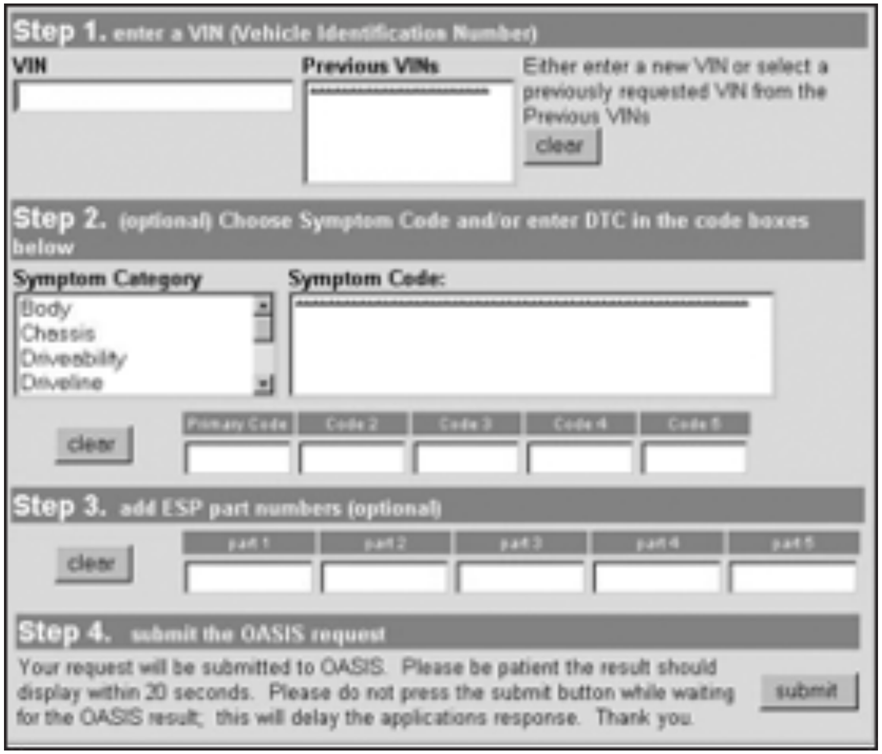
*Dealer-funded components.

Oasis Report

The program website lets the dealership print OASIS Reports for all Ford Brand Ford Blue Advantage® certification-eligible vehicles.

Under the Admin tab, clicking on the OASIS Report item will open the screen below.

- Fill in the information in the available fields as shown below and click Submit



- This will produce an OASIS Report for the desired vehicle as shown in the example below

- Notate any unrepaired recalls on the Ford Blue Advantage Delivery Checklist
- The customer must initial the recall notification on the Ford Blue Advantage Delivery Checklist

Vehicle History

- The program website lets enrolled dealerships check the CARFAX® database of Ford and competitive make vehicles for any vehicle history events that would disqualify the vehicle from the program as well as check for any open safety, compliance, or emission field service actions. This process is designed to help you decide whether a vehicle is eligible for the CPO program.
- This module can be accessed by clicking Vehicle Eligibility Check from the Admin tab drop-down menu

— Select Admin tab on homepage

— Click Vehicle Eligibility Check

— Enter VIN and vehicle mileage; click Continue

— System will advise if the vehicle is eligible to be certified

CARFAX Vehicle History Reports are also provided for all vehicles certified as Ford Blue Advantage® Gold and Blue at no additional charge:

- CARFAX Advantage Dealers can run CARFAX reports for Ford Blue Advantage and Lincoln CPO vehicles at no additional cost
- Dealers on a CARFAX pay-per-VIN pricing plan can request reports at no additional charge – although you may see a charge for the VIN and then a corresponding credit on your bill from CARFAX
- Ford Blue Advantage and Lincoln CPO enrolled dealers without a CARFAX package can run a report free of charge on Ford Blue Advantage and Lincoln CPO vehicles

Program policy requires that a CARFAX Vehicle History Report be run for all vehicles to be certified and that a copy be provided to the customer at delivery and a copy retained in the deal jacket for that vehicle.

- This capability to print the report is part of the certification registration process to be covered under the explanation of that menu item

Note: Dealers must provide their customer with a CARFAX Vehicle History Report and ensure that the CARFAX report has no adverse information that would disqualify a vehicle from being certified under the Ford Blue Advantage program guidelines.

Printing Vehicle Inspection Checklists

- The program website lets you print many of the forms needed for the programs on-demand in the dealership, including the Vehicle Inspection Checklist
- To access this capability, click on the Program Forms item on the drop-down menu, displayed by rolling your cursor over the Marketing Materials tab on the program website homepage
- Roll your cursor over the Marketing Materials tab and then over the Forms item. A selection window will appear to the side of the Forms item showing Ford or Lincoln
- Click on the desired form to open a separate window from which the form can be printed
- To print the desired form, select Print from the File menu. When finished, close the window

Section 2

Using the Vehicle Inspection Checklists

Inspection Checklist Use

Using the Vehicle Inspection Checklist

The Vehicle Inspection Checklists are to be used by the eligible technician to inspect vehicles for certification under the Ford Blue Advantage® – Gold Certification program.

After inspecting and evaluating each checklist item, the technician selects one of the following items that describes the checklist component's condition or how it was resolved:

☐ Passed

☐ Repaired

☐ Replaced

☐ Not Applicable

Major inspection categories on the Ford Blue Advantage Gold and Blue Certification checklists include:

- Vehicle History
- Road Test Performance
- Vehicle Exterior
- Vehicle Interior
- Vehicle Diagnostics
- Underhood
- Hybrid/Electric
- Underbody
- Convenience

All Vehicle Inspection Checklists are available for in-dealership printing from the program website, or printed NCR versions may be ordered at no charge from the Dealer eStore.

Technicians can also complete the Ford Blue Advantage Certification Inspection online using the digital certification application available on the Ford Blue Advantage program website.

Reference copies of the Ford Blue Advantage Vehicle Inspection Checklists are included in the Appendix of the Sales, Service, and Operations guide.

Any open recall(s) must be noted with the FSA/Recall number and description under Step 1: Service Recalls (OASIS) Performed.

With the provision that all of the appropriate steps for a vehicle are completed, individual dealerships may reorganize/reorder specific steps as desired.

Section 3

Gold Certified 172-Point Inspection Process Details

Gold Certified
172-Point Inspection Process Details

The Ford Blue Advantage® — Gold Certification Inspection Checklist contains up to 172 points. Not all points will be applicable to every vehicle. The items in this guide are organized in a logical sequence to make their completion as efficient as possible.

Any open recall(s) must be noted with the FSA number and description under Step 1: Service Recalls (OASIS) Performed.

With the provision that all of the appropriate steps for a vehicle are completed, individual dealerships may reorganize/reorder specific steps as desired.

The Ford Blue Advantage Gold Certification 172-Point Inspection Checklists include the following updates from the prior Ford Certified Pre-Owned 172-Point Inspection Checklists:

- #72 – Integrated Child Safety Seats: Removed
- #76 – Remote Entry System and Push-Button Start System: Split into two Points - #76 and #77
- #171 – Universal Transmitter (Garage Door Opener): Moved to #63

NOTE: All requirements for the Ford Blue Advantage – Gold Certification 172-Point Inspection Checklist are unchanged from the prior Ford Certified Pre-Owned 172-Point Inspection Checklists.

Part One: Vehicle History

Step 1: Service Recalls (OASIS) Performed

Using the Ford Blue Advantage inventory certification module and OASIS, check for and complete all open recalls on the subject vehicle and make any necessary repairs, including software updates for a hybrid vehicle that have not been performed before placing the vehicle into Ford Blue Advantage inventory.

- The Ford Blue Advantage inventory certification module will instantly check for open recalls when a VIN is entered and will provide dealers with immediate feedback
- Dealers should always complete this step of the certification process
- In the event a dealer cannot complete an open recall because of parts availability, annotate the checklist in the comments section, order appropriate parts, and schedule the vehicle repair for future completion
- Should the vehicle be sold prior to parts becoming available:
  - Notate on the Ford Blue Advantage Delivery Checklist the FSA number and description for the open recall.
  - The customer must initial the open recall notation on the Ford Blue Advantage Delivery Checklist
  - Provide the customer a copy of the FSA Sample Owner Letter or if unavailable, the Advance Notice Bulletin.
  - Contact the customer when parts become available to schedule a service appointment for the repair per Ford Motor Company Warranty guidelines

Print a copy of the OASIS Report and place it in the service file. Federal law requires dealers to complete safety recall service before delivering a new vehicle to a buyer or lessee. Although this federal law does not apply to used or certified pre-owned vehicles, dealers should check OASIS for open safety recalls and perform recall service on all Ford products that come into dealerships for service or as sales inventory per the Ford Blue Advantage program guidelines. Before delivering a used or Ford Blue Advantage Certified vehicle with an open safety recall (e.g., parts are unavailable), dealers should consult with their attorneys regarding any regulations and liability laws that may apply in their particular jurisdiction and circumstances. The dealer assumes all risk for any vehicles sold with open safety, emissions, and/or compliance recalls.

Part One: Vehicle History

Step 2: VIN Inspection

Why?

- To ensure that VINs match on the vehicle and corresponding paperwork
- To ensure that recall updates have been performed
- To ensure that California-spec vehicles are equipped with a California emission sticker
- To make certain that scheduled maintenance has been performed
- To make certain that the vehicle is not reacquired

How?

1. Check to see that the VIN plate is properly attached, original, and without alterations
2. Check to see that the VIN plate matches other VINs on the vehicle
3. Check to see that the VIN plate matches corresponding paperwork in the vehicle folder

Step 3: Vehicle History Report Obtained

Using the program website, request a Vehicle History Report to verify the vehicle is, in fact, eligible for certification. See the Vehicle Eligibility section of this guide for a list of conditions/circumstances that would cause a vehicle to be ineligible for certification under the Ford Blue Advantage® – Gold Certification Program.

A copy of the Vehicle History Report should stay with the vehicle so Sales Consultants can use it later when presenting the vehicle to a customer. A copy of the History Overview report is also to be provided to the customer at delivery and a copy retained in the deal jacket.

Step 4: Scheduled Maintenance Performed

Check vehicle records to see that all scheduled maintenance is up to date. Perform scheduled maintenance as necessary.

Step 5: Vehicle Emissions Sticker (Applicable States)

Vehicles equipped with California-spec emissions equipment must also have a California emission sticker in place. Vehicles in other states that also require California-spec emissions must also have the California emission sticker in place.

Part Two: Road Test

Step 6: Engine Starts and Idles Properly

Why?

- To gauge cold and hot engine starts, ruling out ignition, fuel, or air intake malfunctions
- To evaluate the engine's idle quality after cold starts and extended operation

How?

1. With the engine cold, start the vehicle and make sure that the engine starts without excessive cranking. Check that the engine reaches fast idle immediately; evaluate the idle quality for smoothness
2. Following the test-drive, shut off the engine, perform other inspections for approximately 10 minutes to "heat soak" the engine and restart it, checking for excessive cranking time and evaluating the idle quality for smoothness
3. Be alert to:
   - Engine backfiring
   - Long cranking times
   - Loose or cracked vacuum lines if engine idle is rough

Conditions Requiring Repair:

- Failure to start at any time
- Extended cranking times for either cold or hot starts
- Excessively fast or slow idle
- Rough idle

Part Two: Road Test

Step 7: Remote Start System Operation

Why?

- To verify that the Ford Remote Start System is operating properly and starts the engine from outside the vehicle using the remote control

How?

1. Verify the operation of the remote control buttons
2. Test the Remote Start System and verify proper operation as indicated by the appropriate service materials or Owner's Manual

Conditions Requiring Repair:

- Inoperative remote control. Check and replace the battery if necessary. Consult the Owner's Manual for battery information
- Vehicle does not start. Inoperative or malfunctioning Ford Remote Start System. Refer to the appropriate service materials for additional information

Step 8: Engine Accelerates and Cruises Properly/Smoothly

Why?

- To evaluate the engine's performance under light, moderate, and hard acceleration
- To evaluate the engine's performance under slow, moderate, and highway cruising conditions

How?

1. Test-drive the vehicle at various speeds
2. Record any engine performance problem
3. Be alert to:
    - Exhaust smoke under acceleration/deceleration
    - Engine stalling/hesitation/surging at cruise
    - Rough engine idle at stops
    - Engine noise/spark knock (detonation)

Conditions Requiring Repair:

- Rough idle/stalling
- Excessively fast or slow idle
- Hesitation under acceleration
- Blue or white exhaust smoke
- Unusual noises not associated with normal engine operation

Step 9: Engine Noise Normal (Cold/Hot and High/Low Speeds)

Why?

- To ensure that engine noise is normal under all operating conditions

How?

1. Test-drive the vehicle with the engine cold and evaluate engine noise at various speeds during both cold and hot engine operation; record any engine noise conditions
2. Be alert to:
    - Valve lifter/follower ticking or clatter
    - Main and rod bearing noises
    - Engine accessory noise
    - Loose exhaust hanger(s)

Conditions Requiring Repair:

- Unusual noises not associated with normal engine operation

Part Two: Road Test

Step 10: Auto/Manual Transmission/Transaxle Operation – Cold and Hot Shift Quality

Why?

- To evaluate the operation/shifting of automatic and manual transmissions/ transaxles during cold and hot operation

How?

Automatic Transmissions/Transaxles:

1. Slowly place the transmission into each gear throughout the shifter range
2. Listen for unusual sounds, such as excessive driveline clunking, whining, etc.
3. Evaluate the transmission/transaxle operation and shift quality under acceleration and at various speeds during both cold and hot operation

Conditions Requiring Repair:

- Hard or prolonged shifts
- Slippage
- Missing shifts
- Failure to shift into gear
- Worn components

Manual Transmissions/Transaxles:

1. Slowly place the shifter into all gears
2. Listen for any unusual sounds, such as grinding metal
3. Feel for excessive resistance when placing the shifter into each gear

Conditions Requiring Repair:

- Failure to shift into gear
- Worn components

Step 11: Auto/Manual Transmission/Transaxle Noise Normal – Cold and Hot

Why?

- To see if any unusual noises are present during cold and hot operation

How?

Automatic Transmissions/Transaxles:

1. Slowly place the transmission into each gear throughout the shifter range
2. Listen for unusual sounds, such as excessive driveline clunking, whining, etc.
3. Under both cold and hot operating conditions, listen for unusual sounds (whining, whirring, grinding, clanging, etc .) during acceleration/ deceleration and cruise conditions at both low and high speeds

Manual Transmissions/Transaxles:

1. Listen for unusual noises, such as metal grinding
2. Note shifting difficulties or the metal whir of a worn throwout bearing
3. Note the location of any unusual sounds heard while testing the transmission

Conditions Requiring Repair:

- Abnormal transmission noises
- Worn components

Step 12: Shift Interlock Operates Properly

Why?

- To ensure that automatic transmission/transaxle-equipped vehicles shift out of Park only when the brake pedal is depressed

How?

1. With the ignition key in the “ON” position, gently attempt to move the gear selector out of the Park position without applying the brake pedal (it should not move out of position)
2. With the ignition key in the “ON” position, apply the brake pedal and move the gear selector out of the Park position

Conditions Requiring Repair:

- The vehicle shifts out of the Park position without applying the brake pedal first
- The vehicle will not shift out of the Park position with the brake pedal applied

Part Two: Road Test

Step 13: Drive Axle/Transfer Case Operation Noise Normal

Why?

- To listen for unusual bearing/gear noise or unit vibration

How?

1. Evaluate the drive axle and/or transfer case by listening for whining or whirring sounds or unit vibration at low and high speeds, during both cold and hot operation
2. Four-wheel-drive vehicles equipped with transfer case: test-drive in both 2WD and 4WD modes; during 4WD operation, test in both 4-LO and 4-HI transfer case gear ranges; return to 2WD operation and evaluate to ensure hubs are properly disengaged
3. Note any special conditions and/or speed ranges in which bearing/gear noise or vibration is detected, to help the technician make proper repairs

Conditions Requiring Repair:

- Failure to engage/disengage
- Drive axle or transfer case noise caused by worn bearings, bushings, or gears
- Due to worn or broken axle/transfer case bushings, mounts
- Vibration due to component imbalance

Step 14: Clutch Operates Properly

Why?

- To ensure that the clutch operates smoothly and is properly adjusted (if equipped)

How?

1. Depress the clutch pedal; note any unusual or uneven clutch pedal pressure
2. Slowly place the shifter into all gears
3. Listen for any unusual sounds, such as grinding metal, and feel for resistance
4. Note very abrupt or excessively slow clutch take-up
5. Observe how the transmission/transaxle reacts under acceleration; note any clutch slippage or burning
6. Note shifting difficulties or the whirring sound of a worn throwout bearing

Conditions Requiring Repair:

- Worn components
- Clutch slippage/burning
- Grinding metal when placing shifter into gear
- Improper adjustment

Step 15: Steers Normally (Response, Centering, and Free Play)

Why?

- To verify that the steering system operates safely and correctly

Why?

1. Make a variety of turns at various vehicle speeds
2. Evaluate vehicle tracking on a flat road (crowned roads affect straight-line tracking)
3. Evaluate steering wheel for ease/smoothness of operation, free play, ability to readily return to center after a turn, and correct clocking of the steering wheel with the vehicle's wheels in the straight-ahead position
4. To assist the repairing technician, note any roughness in steering action, excessive free play, reluctance of the steering wheel to return to center, pulling to the left or right (check tire inflation when vehicle pulls), and steering wheel clocking

Conditions Requiring Repair:

- Vehicle pulls to the left or right with tires properly inflated
- Excessive free play
- Improper clocking of the steering wheel position
- Steering wheel doesn't readily return to center after a turn
- Steering effort is excessively heavy

Part Two: Road Test

Step 16: Body and Suspension Squeaks and Rattles

Why?

- To inspect for loose or worn body and suspension components

How?

1. Drive the vehicle over a moderately bumpy road and listen for squeaks, rattles, or groans
2. Pinpoint the conditions that cause a squeak or rattle and note where the noise is coming from, such as a loose exhaust hanger, squeaking bushings, etc .
3. Have an assistant drive the vehicle in small circles around you to listen for wheel assembly noises
4. Pinpoint any noted noises during the under-vehicle inspection, when the vehicle is on the hoist

Conditions Requiring Repair:

- Body and/or suspension noises caused by loose or worn parts

Step 17: Struts/Shocks Operate Properly

Why?

- To ensure that struts and/or shocks are properly damping road flaws

How?

1. Place both hands at one corner of the vehicle
2. Rock the vehicle up and down a few times and watch the strut/shock absorber action
3. Repeat the process at each corner of the vehicle
4. If any corner of the vehicle bounces up and down excessively, the strut or shock may be worn or damaged
5. If the vehicle is equipped with automatic load leveling, refer to the Service Manual for the inspection process

Conditions Requiring Repair:

- Replace struts/shocks that allow a corner of the vehicle to bounce up and down excessively; refer to Step 160 for further diagnosis
- For vehicles equipped with automatic load leveling, diagnose and replace components as directed in the Service Manual

Step 18: Brakes/ABS Operate Properly

Why?

- To check the safety and proper operation of the braking and ABS
- To check for appropriate pedal effort and height and for pulsation under moderate braking (Note: ABS may pulsate under hard braking)
- To identify possible steering problems connected to the braking system, such as the vehicle pulling to one side under braking

How?

1. In an isolated area of the dealership lot, accelerate the vehicle to 30 mph and quickly apply the brakes; evaluate for appropriate pedal effort and height
2. Lightly hold the steering wheel in both hands
3. Evaluate the vehicle's braking characteristics and ability
4. Listen for screeching or metal-to-metal sounds when braking
5. If you have applied enough sudden pressure to the brake pedal, ABS will engage
6. When ABS engages, you will feel the brake pedal pulsing and hear a noise from the front of the vehicle; this is normal
7. If the vehicle pulled to one side during braking, note which side to help identify alignment, brake component, and/or tire problems
8. Make another moderate stop and evaluate for pulsation that may indicate warped brake rotors or drums

Conditions Requiring Repair::

- Metallic screeching noises during braking indicate severe brake pad/shoe wear, and that the brake system should be serviced, possibly requiring brake rotor/drum replacement as well
- The brake pedal should be firm (not "squishy") and should not travel too closely to the floor under pressure (either condition requires brake hydraulic system service)
- If the brake warning light illuminates, it has identified a problem in the brake hydraulic system, which will require servicing
- If the ABS warning light remains on or comes on without the brake pedal applied while driving, the braking system should be serviced immediately using the appropriate service publication
- Brake pedal pulsation under moderate braking should be noted and further diagnosed

Part Two: Road Test

Step 19: Cruise Control

Why?

- To check the proper operation and safety of the cruise control unit

How?

1. If you are unfamiliar with the cruise control operation on a specific vehicle, consult the Owner's Manual
2. Ensure that the unit will hold a constant speed at or above 30 mph
3. Test all control switches to make sure they are functioning:
   - On
   - Off
   - Accel
   - Set
   - Coast
   - Resume
4. With the system on and a constant speed set at cruise, depress and release the brake pedal to ensure that the system disengages
5. For manual transmission/transaxle vehicles, repeat this test for the clutch pedal, depressing and releasing it with a constant speed set at cruise to ensure that the system disengages

Conditions Requiring Repair:

- The cruise control fails to engage or hold a set speed
- The cruise control fails to disengage when the brake pedal (or clutch pedal on manual transmission/transaxle-equipped vehicles) is applied
- When any function of the cruise control does not operate or operates improperly

Step 20: Gauges Operate Properly

Why?

- To check the proper operation and illumination of the speedometer, tachometer, odometer, and other gauges

How?

1. After starting the vehicle, turn on the instrument panel lights and ensure that all gauges are illuminated
2. Make sure that each gauge is in its normal range, with the vehicle idling
3. As the engine warms up, check the gauges for reading changes
4. During the test-drive, check the speedometer and odometer for proper operation and ensure that all other gauges are in their normal ranges

Conditions Requiring Repair:

- Replace any burned-out instrument panel bulbs
- Diagnose and repair any sticking, erratic, or broken gauges

Step 21: Driver Select/Memory Profile Systems

Why?

- To check the proper operation of the Driver Select/Memory Profile systems

How?

1. If you are unfamiliar with Driver Select/Memory Profile system settings and/or features, consult the Owner's Manual for assistance
2. Using the security system remote control key fob transmitter, disarm the security system and observe driver's seat and outside mirrors to determine whether any settings have been made to the system; check to see whether adjustments are being made
3. If no adjustments are being made, move the driver's seat and outside mirrors to other positions; close the door, and arm the security system
4. Disarm the security system and observe the driver's seat/outside mirrors to see whether adjustments are being made
5. Move the driver's seat/outside mirrors to other positions and check each of the in-vehicle memory profile settings to see whether adjustments are being made

Conditions Requiring Repair:

- Failure of the key fob or in-vehicle memory profile command to make changes to the seat and/or mirror settings (make certain that settings are entered into memory)
- Failure of the key fob or in-vehicle memory profile command to duplicate the settings entered into the memory profile (make sure that the settings have been correctly entered into memory)

Part Two: Road Test

Step 22: No Abnormal Wind Noise

Why?

- To ensure that there are no abnormal wind noises present while driving the vehicle

How?

1. Test-drive the vehicle with all accessories off and the windows closed
2. Inspect the seals around the windows and doors for damage
3. Inspect the roof rack, mirrors, and any other external components for any physical damage

Conditions Requiring Repair:

- Leaking or damaged seals or weather strip
- Any damaged external components

Part Three: Vehicle Exterior – Body Panels and Bumpers

Step 23: No Evidence of Flood, Fire, Major, or Hail Damage

Why?

- To check for evidence of flood, hail, fire, or other major damage

How?

1. Inspect for any sign of dampness, rust, mud, or gritty dirt that's left over from a flood
2. Inspect for mold, mildew, or unusual odors
3. Inspect for new or mismatched carpet and/or seats
4. Inspect the cargo area carpet for dirt or heavy staining, or for rust underneath (as possible evidence of flood damage)
5. Inspect for smoke damage and odors
6. Inspect body panels, underbody, and engine compartment. Check for any unusual wear or damage
7. Look for vehicle repaint conditions, such as:
   - Sanding/grinding marks
   - Color mismatch
   - Overspray

Conditions Requiring Repair:

- Vehicles with flood, fire, or other major damage cannot be certified
- Hail-damaged vehicles may only be certified if the following conditions are met:
  1. Vehicle title is not/will not be branded due to hail damage
  2. Repairs to remove hail damage from the vehicle have been completed
  3. Hail damage repairs are disclosed to the consumer
     - Documentation regarding hail damage repairs and consumer disclosure must be retained within the vehicle deal jacket
     - If the vehicle suffers hail damage while in Ford Blue Advantage inventory, a dealer may decide to either repair the vehicle within the above guidelines or remove the vehicle from the program via vehicle decertification

Part Three: Vehicle Exterior – Body Panels and Bumpers

Step 24: Body Panel Inspection

Why?

- To inspect for body damage requiring repair or alignment, mismatched paint from prior repairs, poor paint finish quality, tar or tree sap requiring removal, and/or acid rain or industrial fallout exposure

How?

1. Examine body panels, noting any dents and/or scratches that may require repair or refinishing
2. Look for unacceptable vehicle repaint conditions, such as:
   - Sanding/grinding marks
   - Peeling, mottling, run, sag, or bubbling
   - Color mismatch
   - “Orange peel”
3. Inspect for tar or tree sap requiring removal
4. Inspect for exposure to acid rain or industrial fallout

Conditions Requiring Repair:

- Body panel dents or scratches
- Poor-quality vehicle repaint application
- Mismatched repaint color
- Remove tar or tree sap from paint with an appropriate cleaning agent
- Refer to appropriate service materials to remedy acid rain/industrial fallout exposure

Note: When properly performed, minor body- and paint-work greatly enhances the appearance of a vehicle.

Step 25: Bumper/Fascia Inspection

Why?

- To inspect for damage requiring repair/replacement, mismatched paint from prior repairs, poor paint finish quality, tar or tree sap requiring removal, and/or acid rain or industrial fallout exposure

How?

1. Examine front and rear bumpers, fascia and valance panels, noting any dents, scratches, or other physical damage that may require repair, replacement, or refinishing
2. If applicable, look for unacceptable bumper conditions, such as chipped, flaking, or rusty plating
3. If applicable, look for unacceptable bumper/fascia repaint conditions, such as:
   - Sanding/grinding marks
   - Peeling, mottling, run, sag, or bubbling
   - Color mismatch
   - “Orange peel”
4. Check for bumper/fascia misalignment with body panels
5. Inspect for tar or tree sap requiring removal
6. Inspect for exposure to acid rain or industrial fallout

Conditions Requiring Repair:

- Physical damage or misalignment with body panels
- Poor-quality repaint application
- Mismatched repaint color
- Chipped, flaking, or rusted bumper plating
- Remove tar or tree sap from paint with an appropriate cleaning agent
- Refer to appropriate service materials to remedy acid rain/industrial fallout exposure

Note: When properly performed, minor body- and paint-work greatly enhances the appearance of a vehicle.

Part Three: Vehicle Exterior – Doors, Hood, Decklid/Tailgate

Step 26: Doors, Hood, Decklid/Tailgate, and Roof Inspection

Why?

- To inspect for damage requiring repair/replacement, mismatched paint from prior repairs, poor paint finish quality, tar or tree sap requiring removal, and/or acid rain or industrial fallout exposure

How?

1. Examine doors, hood, decklid/tailgate and roof, noting any dents, scratches, or other physical damage that may require repair, replacement, or refinishing
2. Check each component for misalignment with body panels
3. Inspect for unacceptable repaint conditions, such as:
   - Sanding/grinding marks
   - Peeling, mottling, run, sag, or bubbling
   - Color mismatch
   - "Orange peel"
4. Inspect for tar or tree sap requiring removal
5. Inspect for exposure to acid rain or industrial fallout

Conditions Requiring Repair:

- Physical damage
- Poor-quality repaint application
- Mismatched repaint color
- Remove tar or tree sap from paint with an appropriate cleaning agent
- Refer to appropriate service materials to remedy acid rain/industrial fallout exposure

Note: When properly performed, minor body- and paint-work greatly enhances the appearance of a vehicle.

Step 27: Doors, Hood, Decklid/Tailgate Alignment

Why?

- To inspect for damage requiring repair/replacement, mismatched paint from prior repairs and/or poor paint finish quality

How?

1. Check each component for misalignment with body panels
2. Check jambs for paint overspray from previous repairs

Conditions Requiring Repair:

- Misalignment with other body panels; properly align any component that causes an unattractive panel gap or causes binding when the hood, decklid, tailgate, or doors are opened

Note: When properly performed, minor body- and paint-work greatly enhances the appearance of a vehicle.

Step 28: Automatic/Manual Hood Release Mechanisms, Hinges, Prop Rod/Gas Struts Operate Properly

Why?

- To inspect for damage requiring repair/replacement

How?

1. Check proper operation of automatic and manual release mechanisms
2. Check latching and unlatching of components for proper operation and alignment with body panels
3. Check prop rods/gas struts for damage or worn components

Conditions Requiring Repair:

- Misalignment with other body panels; properly adjust any latches or mechanisms that cause binding when the hood, decklid, tailgate, or doors are opened
- Replace worn or damaged components

Part Three: Vehicle Exterior – Doors, Hood, Decklid/Tailgate

Step 29: Power Liftgate, Power Sliding Door Operation

Why?

- To inspect for damage requiring repair/replacement of power liftgate and power sliding doors (if equipped)

How?

1. Inspect power liftgate, power sliding doors for damage
2. Check proper operation of all power switches, including remote control
3. Check proper operation of power liftgate and power sliding doors, including reverse operation

Conditions Requiring Repair:

- When any function of the power liftgate and power sliding doors does not operate or operates improperly
- Physical damage
- Replace worn or damaged components

Part Three: Vehicle Exterior – Grille, Trim and Roof Rack

Step 30: Grille, Trim, and Roof Rack Inspection

Why?

- To inspect for damage or improper attachment/anchoring of vehicle grille, trim, or roof rack

How?

1. Inspect vehicle grille, trim, badging, and roof rack (if equipped) for damage
2. Ensure that each component is properly attached/anchored to the vehicle
3. Note any cosmetic conditions that need to be corrected, such as peeling or flaking paint, flaking, or tarnished badging, etc.

Conditions Requiring Repair:

- Replace damaged or missing grille, trim, badging or roof rack, if equipped
- Replace or reattach loose or improperly attached components
- Polish dull or tarnished brightwork

Step 31: Deployable Running Boards (If Equipped)

Why?

- To ensure the deployable power running boards operate properly, moving automatically when the doors are opened

How?

1. Open the doors and observe the operation of the power running boards
2. When the power running boards are disabled through the message center, they immediately move to the stowed position regardless of the position of the doors
3. When the power running boards are enabled through the message center, they extend when the doors are opened and return to the stowed position when the doors are closed or vehicle speed is greater than 5 mph (8 km/h)

Conditions Requiring Repair:

- Mechanical or electrical damage to the power running boards. Refer to the appropriate service materials for repair information

Part Three: Vehicle Exterior – Glass and Outside Mirrors

Step 32: Windshield, Side- and Rear-Window Glass Inspection

Why?

- To inspect the windshield and the vehicle's side and rear windows for damage

How?

1. Visually inspect the windshield, side, and rear windows for damage including, but not limited to, the following:
    - Cracks
    - Chips
    - Pits
    - Bull's-eyes
    - Wiper marks

Conditions Requiring Repair:

- Any damage to the windshield glass requires repair or replacement
- Glass repairs may not be made in the driver's line of sight
- Any damage to the side- or rear-window glass requires repair or replacement
- Replace with Ford Motor Company Carlite® glass only

Step 33: Wiper Blade Replacement

Why?

- To ensure clear vision during wet weather and to enhance customer satisfaction
- To replace the wiper blades at this time in compliance with program policy

How?

1. Remove and replace the wiper blades (front and rear, as applicable) with the proper Motorcraft® replacement blades (all Ford Blue Advantage® Certified vehicles must receive new wiper blades as part of the reconditioning process)

Conditions Requiring Repair:

- Replace all wiper blades with Motorcraft wiper blades, regardless of condition

Step 34: Exterior Lights (Back/Side/Front)

Why?

- To inspect the vehicle's outside mirrors for damage
- To inspect electrically heated mirrors on equipped vehicles for proper operation
- To inspect folding mirrors for proper operation
- To inspect the headlamps, parking lights, turn signals, and fog lamps/driving lamps for damage
- To verify that headlamps (including daytime running lamps) are operational (both high and low beams)
- To verify that parking lamps, turn signals, and fog lamps/driving lamps are operational
- To verify that headlamps and fog lamps/driving lamps are properly aimed
- To verify that the lights-on reminder chime operates properly
- To inspect the taillamps, brake lamps, high-mounted brake light, turn signals, backup lamps, and license plate lamps for damage
- To verify that taillamps, brake lamps, high-mounted brake light, turn signals, backup lights, and license plate lamps are operational
- To inspect the side marker lamps and puddle lamps for proper operation
- To inspect the side marker lamps and puddle lamps for damage
- To inspect the hazard lamps (emergency flashers) for proper operation
- To verify that headlamps (including daytime running lamps and nighttime auto-on features, as equipped) are operational

How?

1. Visually inspect the outside rearview mirrors for housing or glass damage
2. Inspect mirror adjustment for proper range of movement and operation
3. Consult the vehicle Service Manual to check electrically heated mirrors on equipped vehicles for proper defrost/defog operation
4. Check mirrors with integral turn signals for proper operation
5. Check folding mirrors for proper operation
6. Visually inspect the front-end exterior light lenses for damage such as stone chips

Part Three: Vehicle Exterior – Glass and Outside Mirrors

Step 34: Exterior Lights (Back/Side/Front), Continued

How?

7. Turn headlamps on to low beam and check for burned-out bulbs
8. Turn headlamps on to high beam and check for burned-out bulbs
9. Check to see whether headlight aiming needs adjustment
10. Check to see whether fog lamps/driving lamps aiming needs adjustment
11. Ensure that the daytime running lamps are operational on equipped vehicles
12. Turn parking lamps and fog lamps/driving lamps on to check for burned-out bulbs
13. With the ignition switch "ON," push the turn signal lever up and down to activate the turn signals. Inspect the exterior turn signal lamps for proper signal operation
14. When test-driving the vehicle, the turn signals should self-cancel (turn off automatically) after completing a turn. If they don't self-cancel, turn the signal off manually and note it on the checklist
15. Visually inspect the rear exterior light lenses for damage such as stone chips
16. Turn on the parking lamps and inspect the taillamps and license plate lamps for burned-out bulbs
17. Vehicles with multiple taillight bulbs in each taillight lens: Ensure that illumination on each side of the vehicle is equal. If one side is dimmer than the other, a bulb may be burned out on the dim side
18. Start the vehicle and engage the parking brake
19. Depress the brake pedal and have an assistant confirm that all brake lamps illuminate (including the rear high-mounted brake light)
20. Vehicles with multiple brake lamps in each taillight lens: Ensure that illumination on each side of the vehicle is equal. If one side is dimmer than the other, a bulb may be burned out on the dim side
21. Depress the brake pedal, place the transmission in reverse, and have an assistant inspect the reverse lamps to ensure that they are lit
22. Turn on the parking lamps and walk around the vehicle to inspect illumination of all side marker light assemblies; look for damaged lenses
23. Open the doors and inspect the illumination of all puddle lamps; look for damaged lenses
24. If you are unable to locate it easily, check the Owner's Manual for the location of the hazard light (emergency flasher) switch
25. Activate the hazard lamps. They should flash whether the vehicle is running or not, and no ignition key should be necessary to activate them
26. Activate the headlamps in the auto mode on equipped vehicles
27. Check that the daytime running lamps or nighttime auto-on feature(s) are operational

Conditions Requiring Repair:

- Any damage to the mirror housing or glass requires repair or replacement
- Inoperative or inadequate mirror adjustment requires repair or replacement
- Inoperative electrically heated mirrors
- Inoperative integral turn signals
- Inoperative folding mirrors
- Damaged lenses or housings
- Inoperative or burned-out headlamps or bulbs
- Inoperative daytime running lamps or nighttime auto-on feature(s) on equipped vehicles
- Poorly aimed headlamps
- Corroded bulb sockets and/or bulb contact points
- Inoperative brake lamp switch
- Burned-out side marker or puddle light bulbs
- Damaged side marker or puddle light lenses
- Inoperative hazard lamps (refer to the Service Manual for possible causes)
- Inoperative auto-on/-off lamps (refer to the Service Manual for possible causes)
- Consult the vehicle Owner's Manual for bulb replacement numbers

Note: Handle halogen headlight bulbs carefully. Grasp the bulb only by its plastic base or with a clean cloth and do not touch the glass. The oil from your fingers can cause the bulb to break the next time the headlamps are turned on.

Part Three: Vehicle Exterior – Glass and Outside Mirrors

Step 35: Trailer Lamp Connector Operation

Why?

- To ensure that the connector supplying power to the trailer lamps (taillamps, brake lamps) is operating properly

How?

1. Verify the exterior lighting system of the vehicle is operating correctly
2. Visually inspect for signs of electrical damage or corrosion to the trailer tow connectors

Conditions Requiring Repair:

- Damaged trailer lamp connector
- Inoperative trailer lamp connector is not receiving power or ground. Refer to the appropriate service materials for repair information

Part Four: Vehicle Interior – Airbags and Safety Belts

Step 36: Airbags

Why?

- To verify that the airbags are intact, that no malfunction codes are set and that the instrument panel light operates properly

How?

1. Inspect the airbags for damage to the cover or evidence of tampering
2. Start the vehicle and watch for the airbags' instrument panel light to illuminate during the test cycle, then shut off after a few seconds
3. If the instrument panel light does not light, remains lit, or flashes, use a diagnostic scan tool to retrieve the malfunction code and record the code on the checklist
4. For vehicles equipped with a passenger-side airbag shutoff feature, verify the key function and that the switch indicator light operates properly

Conditions Requiring Repair:

- Inoperative or burned-out airbag instrument panel light
- Airbag instrument panel light remains lit or flashes
- A malfunction code is set

Part Four: Vehicle Interior – Airbags and Safety Belts

Step 37: Safety Belts

Why?

To verify that the safety belts operate properly and are free from cuts or wear

How?

1. Pull out and inspect the safety belt webbing to check for any cuts, fraying, or wear in the material
2. Check each safety belt buckle for proper operation, ensuring that it locks and releases properly and without difficulty
3. On vehicles with height-adjusting shoulder belts, ensure that the adjusters move freely and are securely attached
4. Check passenger's outboard locking retractors as follows:
   - Pull the safety belt webbing out of the retractor to the extent of its travel
   - Listen for a ratcheting sound while feeding the webbing back into the retractor
   - When this sound is heard, pull the webbing outward
   - If the webbing locks in place immediately, the locking mechanism is operating properly
   - Permit the webbing to fully retract, then verify that the retractor has reset
5. Check the safety belt webbing for an energy-absorbing sew pattern that provides a measure of "give" in the event of a collision. This may be located in a short plastic boot on the front seat outboard anchor location. If the sew pattern has been released, a label or lettering marked "REPLACE BELT" may appear. If this lettering is visible, all outboard belts must be replaced

Conditions Requiring Repair:

- Tears, cuts, fraying, or heavy wear in the safety belt webbing material
- Malfunctioning safety belt buckle(s)
- Malfunctioning passenger's outboard locking retractors
- Safety belts with a released energy-absorbing sew pattern (replace all outboard belts in this instance)

Part Four: Vehicle Interior – Audio and Alarm Systems

Step 38: Radio, Radio Antenna, Cassette, CD, Satellite Radio, and Speakers

Why?

- To verify that the radio, CD player, satellite radio, and speakers on equipped vehicles operate properly with acceptable sound clarity
- To verify proper positioning, condition, and operation of the fixed-mast or power radio antenna

How?

1. For assistance in inspecting the audio system, consult the vehicle Owner's Manual for a full overview of operating features
2. Test each operational feature of the radio, CD player, clock, graphic equalizer, satellite radio, and remote radio controls as they apply to the vehicle being inspected, including steering wheel controls and rear seat controls
3. Listen for and note any sound distortion at reasonable volume levels
4. If the radio is equipped with an anti-theft security code, verify that the code has been cleared and/or disabled
5. If the vehicle is equipped with SiriusXM® Radio, record the ESN # and activate the SiriusXM Dealer Demo Service. Information on how to accomplish this can be obtained by calling SiriusXM Dealer Support at 1-800-852-9696, or download the FREE SiriusXM Dealer App from the App Store® or Google Play™ store, and begin activating radios either before or during the delivery process to ensure all your pre-owned customers drive home listening to SiriusXM
6. Visually inspect the condition of the fixed-mast antenna, making sure that it is not bent and is securely fastened to the base
7. Ensure that the antenna base is securely fastened to the vehicle's body panel
8. Turn the radio on to extend a power antenna and check for any damage to the power antenna's telescoping sections
9. Tune in a local radio station to test antenna operation

Conditions Requiring Repair:

- Missing or damaged components
- Operational failure of any audio component or feature
- Inoperative, distorting, or malfunctioning speakers
- Missing or damaged antenna mast or power antenna telescoping section
- Inoperative power antenna
- Inability to tune in local radio stations (if this condition exists, check the antenna lead at the radio head unit to ensure that it is plugged in)

Step 39: Alarm/Theft-Deterrent System

Why?

To verify that the Ford Motor Company alarm system is operating properly

How?

1. Test the system with the remote and verify proper operation as indicated by the Service Manual or Operations Guide

Conditions Requiring Repair:

- Inoperative Ford Motor Company alarm/theft-deterrent system
- Inoperative remote control

Part Four: Vehicle Interior – Audio and Alarm Systems

Step 40: Navigation System (If Equipped)

Why?

- To ensure the touchscreen Navigation System that is integral to the audio system operates properly

How?

1. For assistance in inspecting the Navigation System, consult the Owner's Manual for a full overview of operating features
2. Check that the navigation map DVD (if equipped) is stored in the audio system
3. Verify that all hard key buttons and touchscreen buttons are operating properly
4. Enter a destination and test-drive the vehicle. Check that the Navigation System provides audible and visual instructions

Conditions Requiring Repair:

- Operational failure of any Navigation System component or feature
- The Navigation System provides on-board diagnostics to assist with diagnosing a concern. To carry out self-diagnostics, press and hold presets three (3) and six (6) simultaneously for three (3) seconds
- Follow the on-screen options for additional diagnosis. Refer to the appropriate service materials for repair information.

Step 41: Air Conditioning System

Why?

- To verify that the air conditioning system is operating properly
- To ensure that all air vent passages (front and rear) are unobstructed, that their directional louvers can be fully adjusted, and that positive shutoff levers are fully functional
- To ensure that each of the air vent passage selections is operating properly

How?

1. Place the temperature valve control in the "COLD" position and engage the air conditioning control to turn on the AC
2. Check to make sure that all airflow selections are providing forced air
3. Ensure that the air conditioning unit is blowing cold air
4. For vehicles equipped with an Electronic Automatic Climate Control System, Electronic Dual Automatic Climate Control System, and Rear Electronic Climate Control System, consult the Owner's Manual for specific operation instructions
5. Cycle through all blower fan speeds and listen for a corresponding change in the sound of the incoming air
6. Check the area between the cowl and the body for leaves, twigs, and other foreign matter
7. Be alert to very musty odors that may indicate mildew on the evaporator core face (objectionable odor may require that the evaporator core be disinfected by a technician)

Conditions Requiring Repair:

- Inoperative, missing or malfunctioning air conditioning controls or vents
- Noisy or inoperative blower fan or blower switch that does not change fan speeds
- AC blows warm air
- AC compressor will not engage/disengage
- Objectionable musty odor from vents (check evaporator core drain for plugging and/or disinfect/deodorize evaporator core face as necessary)

Part Four: Vehicle Interior – Heat/Vent/AC/Defog/Defrost

Step 42: Heating System

Why?

- To verify proper heating system operation
- To ensure that all air vent passages (front and rear) are unobstructed, that their directional louvers can be fully adjusted, and that positive shutoff levers are fully functional
- To ensure that each of the air vent passage selections is operating properly

How?

1. Place the temperature valve control in the "HOT" position and engage the heater control to turn on the heating system
2. Check to make sure that all airflow selections are providing forced air to the corresponding vent
3. With the vehicle at operating temperature, ensure that the heating system is blowing warm air
4. Cycle the temperature valve halfway between the "HOT" and "COLD" positions and note the change in air temperature as blended air is forced through the vents
5. For vehicles equipped with an Electronic Automatic Climate Control System, Electronic Dual Automatic Climate Control System, and Rear Electronic Climate Control System, consult the Owner's Manual for specific operation instructions
6. Cycle through all blower fan speeds and listen for a corresponding change in the sound of the incoming air
7. Note any smell of engine coolant and check for coolant or other moisture leaking into the vehicle near the front footwells

Conditions Requiring Repair:

- Inoperative, missing or malfunctioning heater controls or vents
- Noisy or inoperative blower fan or blower switch that does not change fan speeds
- Heater blows cold air with the engine at operating temperature
- Temperature valve adjustment does not result in a temperature change from the vent air
- Engine coolant leaking into passenger compartment (indicates probability of a heater core leak)

Part Four: Vehicle Interior – Heat/Vent/AC/Defog/Defrost

Step 43: Defog/Defrost

Why?

- To verify proper defog/defrost system operation
- To ensure that the defog air vent passage is unobstructed and operating properly
- To ensure that the electric rear defroster operates properly
- To ensure that the electric heated outside mirrors operate properly

How?

1. Place the temperature valve control in the "HOT" position and engage the defog control to turn on the defog system
2. Check to make sure that all airflow selections are providing forced air to the corresponding vent, making sure to check side-window demisters as well
3. With the vehicle at operating temperature, ensure that the defog system is blowing warm air
4. For vehicles equipped with an Electronic Automatic Climate Control System, Electronic Dual Automatic Climate Control System, and Rear Electronic Climate Control System, consult the Owner's Manual for specific operation instructions
5. Turn on the electric rear defroster and check pilot light operation if equipped:
   - Fill a spray bottle with ice-cold water (the colder, the better)
   - Spray the outside of the rear-window with cold water
   - Activate the electric rear window defroster
   - As the heating grid warms up, the grid lines will evaporate the cold water, leaving a line indicating that electrical current is flowing through the grid
   - Inoperative grids or grid lines downstream of a break in the grid will not evaporate the cold water
6. Turn on the electric heated outside mirrors (consult the Owner's Manual for operating instructions) and check for proper operation

Conditions Requiring Repair:

- Inoperative, missing or malfunctioning defrost/defog controls or vents
- Noisy or inoperative blower fan or blower switch that does not change fan speeds
- Defog function blows cold air with the engine at operating temperature
- Inoperative electric rear defroster or pilot light (heating grids that have breaks in them are repairable; see the appropriate Service Bulletin for procedures)
- Inoperative electric-heated outside mirrors

Part Four: Vehicle Interior – Interior Amenities

Step 44: Clock

Why?

- To check operation and accuracy of instrument panel clock on vehicles with a clock separate from the audio system

How?

1. Inspect the clock for any physical damage, such as a cracked lens or broken/loose hands
2. Check to see whether instrument panel clock (if equipped) is accurately set; synchronize with your watch or the shop clock as necessary (consult the Owner's Manual for setting procedures)
3. Perform the remainder of the vehicle inspection and recheck the clock against your watch or the shop clock for accuracy

Conditions Requiring Repair:

- A cracked lens or broken/loose hands
- Inoperative clock
- Inability to keep accurate time

Step 45: Tilt/Telescopic Steering Wheel

Why?

- To verify that the tilt/telescopic steering wheel moves through the range of positions, including memory systems, if equipped
- To verify that the tilt steering wheel will lock securely in each position

How?

1. Pull the steering column release lever toward you and move the steering column into each position
2. Make sure that movement is smooth, without binding or looseness in the column
3. For telescopic steering wheels, activate the release lever and pull the steering wheel toward you, testing for smoothness of operation and full range of operation
4. Check all functions of power tilt systems; see the Owner's Manual for operation instructions

Conditions Requiring Repair:

- Difficulty in moving throughout the range of motion, looseness, or binding requires further diagnosis

Note: Never adjust the steering wheel or column while the vehicle is moving.

Step 46: Steering Column Lock

Why?

- To verify that the steering wheel locks securely in place with the ignition turned off

How?

1. Turn the ignition to the "OFF/LOCK" position and remove the key
2. Check that the steering wheel locks in place and cannot be turned

Conditions Requiring Repair:

- Movement of the steering wheel with the ignition turned off requires further diagnosis

Step 47: Steering Wheel Controls

Why?

- To verify that all steering wheel controls (cruise control and audio system, if equipped) are operational and intact

How?

1. Inspect cruise control buttons to ensure that they are not damaged; operation is inspected in Step 19
2. Inspect audio system control buttons to ensure that they are not damaged; operation is inspected in Step 38

Conditions Requiring Repair:

- Broken, cracked, missing, or inoperative controls

Part Four: Vehicle Interior – Interior Amenities

Step 48: Horn, Warning Chimes

Why?

- To verify that the vehicle's steering wheel-mounted horn pad/button operates properly
- To verify horn sound quality
- To verify that the vehicle's ignition key, safety belt, and headlights-on warning chimes are functioning

How?

1. Press the steering wheel-mounted horn pad/button and verify horn operation; it should sound steadily
2. Verify that the horn operates with minimal pressure on the horn pad/button
3. Sitting in the driver's seat with all doors closed, switch the ignition key to "ON" and listen for the safety belt warning chime. While the chime is sounding, buckle the safety belt; the chime should stop sounding
4. Switch the engine off (leaving the ignition keys in ignition) and open the door. The ignition key warning chime should sound; remove the keys and the chime should stop sounding
5. Close the door and switch on the headlights. Open the door and listen for the headlights-on warning chime; with the door open, turn off the headlights, and the chime should stop sounding

Conditions Requiring Repair:

- Excessive pressure required to operate the horn
- Horn does not sound steadily when the pad/button is pressed
- Horn fails to sound when the pad/button is pressed
- An inoperative horn can be caused by a blown fuse, an inoperative horn relay, broken/corroded steering wheel buttons, open electrical wiring circuit, or maybe the fault of the horn unit itself
- Inoperative ignition key, safety belt, and/or headlights-on warning chimes
- A blown fuse may also be the cause of an inoperative chime

Step 49: Instrument Panel and Warning Lights

Why?

- To verify that the vehicle's instrument panel lights, gauges, and warning lights illuminate properly

How?

1. Turn the ignition key to "ON" and observe instrument panel-mounted warning lights for proper illumination
2. Turn the parking lights on and verify that the instrument panel lights illuminate, watching for dark spots on the instrumentation that indicate a burned-out bulb
3. Adjust instrument panel light intensity to check the dimmer function for proper operation

Conditions Requiring Repair:

- Replace any burned-out warning light or instrument panel illumination lights; correct any electrical system conditions as necessary

Part Four: Vehicle Interior – Interior Amenities

Step 50: Wipers, Washers

Why?

- To verify that the vehicle's windshield wipers operate properly
- To verify that the vehicle's rear-window wiper (if equipped) operates properly
- To visually check to see that the wiper arms cycle properly
- To verify that the vehicle's windshield washers operate properly
- To verify that the vehicle's rear-window washer (if equipped) operates properly

How?

1. Turn the ignition key to "ON" or to the accessory position ("ACC") and test the following wiper settings for proper operation:
   - HI/LO wiper speed operation
   - Interval wiper speed settings on equipped vehicles
2. Turn the ignition key to "ON" or to the accessory position ("ACC") and test the washer fluid spray volume/pattern to ensure that the wipers operate without streaking or smearing
3. After releasing the washer fluid button/lever, ensure that the wiper arms continue cycling up to four times before returning to the "park" position

Conditions Requiring Repair:

- Damaged or inoperative wiper arms
- Inoperative washer pump(s)
- Leaking washer fluid reservoir(s)
- Inadequate washer fluid volume or misaligned spray pattern

Step 51: Interior Courtesy, Dome, and Map Lights

Why?

- To verify that the vehicle's interior courtesy, dome, and map lights are functioning

How?

1. Turn on each courtesy, dome, and map light in turn and ensure that all bulbs light when switched on
2. Make certain that the dome light operates both from the instrument panel switch and the doorjamb switches
3. Replace any burned-out bulbs

Conditions Requiring Repair:

- Replace all burned-out bulbs
- Keep in mind that a faulty switch or blown fuse may also be the cause of an inoperative light

Part Four: Vehicle Interior – Interior Amenities

Step 52: Manual Outside Rearview Mirrors, Power Outside Rearview Mirrors, Auto-Dimming Rearview Mirror, and Rear View Camera System

Why?

- To inspect the vehicle's mirrors for damage
- To verify that the vehicle's rearview mirror is properly mounted to the windshield
- To verify that the rearview mirror provides a full range of movement
- To verify that the rearview mirror does not shake or vibrate excessively while the vehicle is in motion
- To verify that the outside rearview mirrors and/or their adjustment controls operate properly and provide a full range of movement
- To inspect the power mirrors and auto-dimming rearview mirror (if equipped) for proper operation

How?

1. Visually inspect the rearview mirror to ensure that it is not cracked, broken, or faded and is properly mounted to the windshield; adjust it gently to test the sturdiness of its mounting
2. Check rearview mirror's range of movement and ability to hold a set position
3. With the vehicle in motion, watch the rearview mirror for movement or vibration that interferes with a clear view to the rear
4. Visually inspect the outside rearview mirrors for housing or glass damage
5. Inspect mirror adjustment for proper range of movement and operation. Check ability to hold a setting and for any vibration or shaking while the vehicle is in motion
6. Check the electrically heated mirrors on equipped vehicles for proper operation
7. On equipped vehicles, inspect the auto-dimming rearview mirror and wiring for proper operation. Check the rearview mirror's auto-dimming function by shining a bright light onto the mirror assembly. The mirror should adjust to dim the glare. Refer to the appropriate service materials for additional information
8. Check for obstructed forward-facing and rearward-facing photoelectric sensors

Conditions Requiring Repair:

- Any damage to the mirror housing or glass
- Poor anchoring of mirror to windshield
- Inoperative or inadequate mirror adjustment
- Inoperative electrically heated mirrors
- Inoperative auto-dimming mirror function (if equipped)
- Any mirror shaking or vibrating excessively with the vehicle in motion

Step 53: Blind Spot Information System (BLIS®)

Why?

- To verify the Blind Spot Information System (BLIS) functions properly

How?

1. Check for proper BLIS operation; refer to workshop manual procedures

Conditions Requiring Repair:

Inoperative or malfunctioning BLIS indicators

Step 54: Rear View Camera System

Why?

- To ensure proper operation of the Rear View Camera System when the transmission is in Reverse

How?

1. Visually inspect the camera on the rear of the vehicle for signs of damage
2. Inspect the inside rearview mirror and wiring for proper operation. Refer to the appropriate service materials for additional information

Conditions Requiring Repair:

- Inoperative Rear View Camera
- Poor image quality

Part Four: Vehicle Interior – Interior Amenities

Step 55: SYNC® System

Why?

- To verify that the SYNC system operates properly

How?

1. For assistance in inspecting the SYNC system, consult the Owner's Manual for a full overview of operating features
2. Test each operational feature of the system
3. Check for and perform necessary upgrades
4. Verify the operation of the steering wheel-mounted controls
5. Test and perform a master reset. Consult the Owner's Manual for the reset procedure

Conditions Requiring Repair:

Operational failure of any system component or feature Inoperative or malfunctioning microphone or speakers

Step 56: MyFord Touch®/Master Reset

Why?

- To verify that the MyFord Touch System operates properly

How?

1. For assistance in inspecting the MyFord Touch System, consult the Owner's Manual for a full overview of operating features
2. Test each operational feature of the system
3. Ensure that vehicles equipped with MyFord Touch technology contain the latest level of software available for that vehicle
4. Test and perform a master reset. Consult the Owner's Manual for the reset procedure

Conditions Requiring Repair:

- Operational failure of any system component or feature

Step 57: Active Park Assist (If Equipped)

Why?

- To verify the Active Park Assist feature operates properly

How?

1. Refer to the Owner's Manual for a full overview of the Active Park Assist feature
2. Test functionality of the Active Park Assist switch
3. Verify operation of Active Park Assist functions

Conditions Requiring Repair:

Inoperative switch or system malfunction preventing Active Park Assist from functioning

Step 58: Rear Entertainment System (If Equipped)

Why?

- To ensure that the rear entertainment system is operating properly

How?

1. For assistance in inspecting the rear entertainment system, consult the Owner's Manual for a full overview of operating features
2. Test each operational feature of the system, including the rear seat controls and A/V ports
3. Verify the operation of the remote control
4. Verify the operation of the wireless headphones
5. Insert a DVD into the player and verify video screen operation
6. Verify single/dual-play operation of the system

Conditions Requiring Repair:

- Operational failure of any system component or feature
- Inoperative or malfunctioning speakers
- Inoperative or malfunctioning video screen
- Inoperative or malfunctioning remote control
- Inoperative or malfunctioning wireless headphones

Part Four: Vehicle Interior – Interior Amenities

Step 59: Power Outlets and Lighter

Why?

- To verify that power is available to all lighters and/or auxiliary 12V power outlets
- To ensure proper operation of the 110V power outlet

How?

1. Push in the lighter element knob and verify that the genuine Motorcraft® lighter pops out into the "ready" position; check the heating element to see whether it is glowing
2. Use a test light to verify that power is available at all auxiliary 12V power outlets
3. If power is not available at any outlet, check the fuse panel for blown fuses; replace fuse and retest
4. With the key in the "ON/START" position, inspect the green LED at the power outlet. This should be illuminated
5. Refer to the appropriate service materials for further diagnosis

Conditions Requiring Repair:

- Nonoriginal equipment lighter knob/element assembly
- Missing lighter knob/element assembly in vehicles originally equipped with the assembly
- Blown fuses may require additional diagnosis
- Inoperative lighter knob/element assembly
- Inoperative lighter socket or auxiliary 12V power outlets may require further diagnosis
- The LED is not illuminated or is flashing with the key in the "ON/START" position
- The LED normally flashes/pulses when the key is in the "OFF" position

Note: Do not plug electrical accessories into the lighter when auxiliary 12V power outlets are available. Electrical system damage could occur.

Step 60: Ashtrays

Why?

- To verify that the ashtray(s) on equipped models are structurally sound and free of cracks
- To verify that all components of the ashtray are intact, including the ash bucket
- To verify that the ashtray doors open and close smoothly

How?

1. Visually inspect the ashtray on equipped models to make sure that all necessary pieces are present
2. Look for broken, bent, or melted components that require repair or replacement

Conditions Requiring Repair:

- Repair or replace missing or damaged components

Part Four: Vehicle Interior – Interior Amenities

Step 61: Glove Box and Center Armrest/Console

Why?

- To verify that the glove box door operates properly and that the glove box light (if equipped) is operational
- To verify that the center armrest operates properly and that the console door (on console-equipped vehicles) is operational

How?

1. Inspect the glove box door and glove box interior for cracks or other damage
2. Check operation of the glove box door latch, lock (if equipped), and hinges for proper fit and operation
3. Locate and verify that the vehicle Owner's Manual and related materials are correct for the vehicle year and model
4. Ensure that the glove box light illuminates when the glove box door is opened
5. Operate the armrest through its range of motion to ensure that it is operational
6. Check the console door, latch, and lock (if equipped) for proper alignment, engagement, and locking action
7. Place flip armrest/console combination on equipped vehicles in vertical and horizontal positions to ensure that it operates properly

Conditions Requiring Repair:

- Repair or replace missing, damaged, or malfunctioning components
- Replace burned-out glove box lightbulb as necessary

Step 62: Sun Visors, Vanity Mirror, and Light

Why?

- To ensure that the sun visors operate properly throughout their entire range of movement
- To ensure proper operation of the vanity mirror and light on equipped models

How?

1. Operate each sun visor through its range of motion and carefully unsnap the inboard clip from its anchor to ensure that the visor will rotate toward the side window as well
2. Check operation of any secondary sun visors by moving them through their range of motion
3. Ensure that the vanity mirror reflective surface is not cracked, chipped, scratched, or faded
4. If equipped, the mirror's flip-up cover and any framing trim around the mirror should not be cracked or chipped
5. If equipped, check the vanity mirror light to ensure that all lights illuminate and that all light lenses are in good condition

Conditions Requiring Repair:

- Repair or replace missing, damaged, or malfunctioning components
- Replace any burned-out vanity lightbulbs as necessary
- Blown fuses are another possible cause of a vanity lightbulb that does not illuminate

Step 63: Universal Transmitter (Garage Door Opener, If Equipped)

Why?

- To ensure that all buttons of the universal transmitter (garage door opener) have been cleared

How?

1. Turn the key to the "ON" position
2. Prepare for programming the universal transmitter by erasing all three channels by holding down the two outside buttons until the red light begins to flash (20–30 seconds). Release both buttons
3. Consult the vehicle Owner's Manual for additional operating instructions about the universal transmitter

Conditions Requiring Repair:

- All buttons of the universal transmitter have not been cleared

Part Four: Vehicle Interior – Interior Amenities

Step 64: Adjustable Pedals (If Equipped)

Why?

- To ensure that the adjustable pedals on equipped vehicles move through their range of motion and adjust properly

How?

1. For full details on adjusting pedals, consult the vehicle Owner's Manual
2. Check operation and adjustability of adjustable pedals by moving them through their range of motion
3. Ensure that the pedal will securely hold an adjustment once it is made
4. Check pedal pads for excessive wear

Conditions Requiring Repair:

- Repair or replace missing, damaged, or malfunctioning components; consult the Service Manual for procedures
- Pedals that will not hold an adjustment
- Pedals that will not move throughout the full range of adjustment
- Worn pedal pads

Note: This is a vital safety item. Closely follow Service Manual procedures when repairs are performed.

Step 65: Interior Odor-Free

Why?

- To ensure that the vehicle interior is clean and odor-free

How?

1. Visually inspect the vehicle's carpeting and upholstery
2. Note areas with heavy staining that may require application of chemical treatment prior to cleaning
3. Feel the carpet for wetness/dampness and lift a corner to inspect the floorpan for rust due to flood damage

Conditions Requiring Repair:

- Strong odors may require complete interior cleaning
- Heavy staining and odors may require carpet or upholstery replacement

Part Four: Vehicle Interior – Carpet, Mats, and Trim

Step 66: Carpet, Floor Mats

Why?

- To ensure that all carpeting is clean and free from wear marks, burn marks, and stains
- To check for flood damage as evidenced by rust beneath the carpet
- To ensure that floor mats (if equipped) are clean and free from wear marks, burn marks ,and stains

How?

1. Visually inspect the vehicle's carpeting
2. Note wear marks, burn marks, and areas with heavy staining that may require application of chemical treatment prior to cleaning
3. Feel the carpet for wetness/dampness and lift a corner to inspect the floorpan for rust due to flood damage
4. Visually inspect the vehicle's floor mats

Conditions Requiring Repair:

- Carpet with heavy staining may not come clean
- Carpet burns and wear spots can be difficult and expensive to repair
- Heavy staining, wear, and burn holes may require carpet replacement
- Vehicles with flood damage cannot be certified
- Floor mats with heavy staining may not come clean
- Heavy staining, wear, and burn holes may require floor mat replacement

Step 67: Door Trim and Door Panels

Why?

- To verify that all door trim, door panels, armrests, and pull straps are securely attached

How?

1. Inspect the fit, finish, and condition of each door trim panel, armrest, and strap
2. Ensure that they are properly secured and in good condition

Conditions Requiring Repair:

- Loose, missing, inoperative, or damaged components require repair or replacement

Step 68: Headliner

Why?

- To verify proper condition and positioning of the interior headliner

How?

1. Visually inspect the headliner for tears, sagging, water-leak stains near the moonroof location (if equipped), burns, and other cosmetic blemishes

Conditions Requiring Repair:

- Replace the headliner if it is unsightly and cannot be cleaned or repaired

Part Four: Vehicle Interior – Seats

Step 69: Seat Upholstery

Why?

- To verify that the front- and rear-seat fabric is clean (not stained, worn, cut, or cracked)

How?

1. Visually inspect the front and rear seat upholstery for stains, burns, excessive wear, cuts, tears, or cracks

Conditions Requiring Repair:

- Repair or replace any component that shows excessive signs of wear
- Replace any component that is not readily reconditioned

Step 70: Seat and Head Restraint Adjustment

Why?

- To ensure that manual and power seats can be adjusted and hold their set positions
- To ensure that head restraints can be adjusted and hold their set positions

How?

1. Operate the front seat(s) throughout their range of movement, including:
   - Fore and aft adjustment
   - Height adjustment (if equipped)
   - Seat back angle adjustment (rake)
   - Seat cushion angle adjustment (if equipped)
   - Lumbar support operation (if equipped)
2. On models equipped with manual seats, make certain that the seat track locks securely into position
3. On models equipped with two-way adjustable head restraints, adjust the head restraints up or down
4. Ensure that the front seat(s) and head restraints will retain their settings throughout the course of the test-drive

Conditions Requiring Repair:

- Seats that will not fully adjust as intended or will not hold a set position
- Head restraints that are broken, missing, or will not hold a set position

Step 71: Folding Seats

Why?

To ensure that manual and power seats (if equipped) can be folded

How?

1. Operate the rear seat(s) throughout their range of movement, including:
   - Fore and aft adjustment (if equipped)
   - Seat back angle adjustment (rake)
   - Seat back folded (if equipped)
   - Seat tumbled forward (if equipped)
2. On models equipped with third-row seats, operate the seat(s) throughout their range of movement, including:
   - Seat back folded (if equipped)
   - Seat folded flat (if equipped)

Conditions Requiring Repair:

- Seats that will not fully adjust or fold as intended or will not hold a set position

Part Four: Vehicle Interior – Seats

Step 72: Heated Seats/Cooled Seats

Why?

- To ensure that heated and/or cooled seats on equipped vehicles operate and adjust properly

How?

1. Operate the control for each heated and/or cooled seat and feel each seat to ensure that it is heating or cooling properly
2. After the seats warm up or cool down, vary the setting and feel for a corresponding change in temperature

Conditions Requiring Repair:

- Seats with inoperative/defective switches, relays, or blown fuses will require further diagnosis and repair

Part Four: Vehicle Interior – Sunroof/Moonroof/Convertible Top

Step 73: Sunroof/Moonroof (If Equipped)

Why?

- To ensure proper sunroof/moonroof operation
- To inspect sunroof/moonroof weather seal integrity

How?

1. Consult the vehicle Owner's Manual for operating instructions for the particular vehicle
2. Fully open and close the sunroof/moonroof
3. Observe how the roof panel travels in the tracks; it should move smoothly
4. Tilt the roof panel up and then close it
5. Visually inspect the top of the roof panel to ensure that it is flush with the vehicle roof surface
6. Inspect wind deflectors for proper operation. Exterior-mounted deflectors should be securely fastened
7. Inspect the tracks in which the roof panel travels; they should be free of dirt and debris
8. Inspect sunroof/moonroof drains for blockage or debris

Conditions Requiring Repair:

- Failure to pass any of the above inspections
- Vehicles with aftermarket sunroofs/moonroofs may be certified if the roof adds value to the vehicle and meets criteria listed above; if an aftermarket roof requires repair, check glove box for roof manufacturer or installer warranty details

Step 74: Convertible Top (If Equipped)

Why?

- To ensure proper convertible top operation
- To inspect convertible top weather seal integrity
- To inspect rear window condition

How?

1. Consult the vehicle Owner's Manual for operating instructions for the particular vehicle
2. Fully open and close the convertible top
3. Observe how the convertible top lowers and rises; it should move smoothly
4. Visually inspect the convertible top for any cuts, stains, faulty weather stripping, or damaged hardware
5. Inspect the condition of the rear window
6. Check the rear window defogger (if equipped) for proper operation

Conditions Requiring Repair:

- Failure to pass any of the above inspections

Part Four: Vehicle Interior – Door Handles, Entry, Locks, and Windows

Step 75: Door Handles and Release Mechanisms

Why?

- To ensure that doors open and close smoothly, fully, and with little effort

How?

1. Inspect the operation of all door handles and cargo hatch/door handles or latches, both inside and out, to confirm sound mechanical operation
2. Inspect all interior and exterior door release handles to ensure that they are not broken, bent, or missing and to ensure that they are securely attached
3. Inspect operation of electrically activated sliding side doors on equipped vehicles by performing the following:
   - Activate the door from inside the vehicle and with the remote control device
   - Check for dirt or debris in the door track that would prevent smooth operation
   - Check the door-close safety feature for safe, proper operation by placing an obstruction such as a roll of paper towels in the doorsill and activating the door; the door should not continue closing after meeting and detecting the obstruction

Conditions Requiring Repair:

- Damaged, missing, or inoperative door handles or cargo hatch/door handles or latches
- Loose, missing, or inoperative components
- Malfunctioning interior switch or remote control on vehicles equipped with electrically activated sliding side doors
- Malfunctioning safety stop on vehicles equipped with electrically activated sliding side doors

Step 76: Remote Entry System

Why?

- To ensure that key fob remote control functions are all operational
- To ensure that door keypads (keyless entry) functions are all operational

How?

1. Consult the vehicle Owner's Manual and related materials for full remote control and keyless entry keypad operating instructions
2. Activate the key fob remote control to ensure that it unlocks the doors and, if equipped, deactivates the alarm/theft-deterrent systems; at least one key-fob remote control should be available
3. Ensure that the keyless entry keypad function operates by entering a new code per the Owner's Manual; test all functions

Conditions Requiring Repair:

- Damaged, missing, inoperative, or poor-performing key-fob remote controls
- Vehicle-mounted keyless entry receiver/activator damaged, missing, inoperative, or performing poorly
- Damaged or inoperative keyless entry keypad

Step 77: Push-Button Start System

Why?

- To ensure that key-fob remote control functions are all operational
- To ensure that push-button start operates properly, if equipped

How?

1. Consult the vehicle Owner's Manual and related materials for full push-button start operating instructions
2. Activate push-button start, if equipped

Conditions Requiring Repair:

- Damaged, missing, inoperative, or poor-performing key-fob remote controls
- Malfunctioning or inoperative push-button start

Part Four: Vehicle Interior – Door Handles, Entry, Locks, and Windows

Step 78: Door Locks

Why?

- To ensure that locks operate properly, and that the door cannot be opened from the outside when the lock button is in the “Lock” position
- To verify that door key locks operate properly
- To verify that power door locks function properly

How?

1. Lock all doors, close them, and attempt to gain entry to the vehicle from the outside at all door or hatch points
2. Use the vehicle key to gain entry to the vehicle wherever a keyhole is present
3. From inside the vehicle, test the power lock function on equipped models and ensure that all doors lock and unlock when the power lock switch is activated in each mode
4. Manually activate each lock knob/button in the Lock and Unlock modes to check for smooth operation

Conditions Requiring Repair:

- Door keyholes do not work
- Power door lock feature is inoperative in Lock, Unlock, or both modes
- Inside door lock knob/button sticks, binds, or does not lock or unlock the door when in position
- Door opens from outside with the lock button in the “Lock” position

Step 79: Child Safety Locks

Why?

- To ensure that the rear doors will not open from the inside with the child safety locks activated on equipped vehicles

How?

1. For full operating instructions, consult the vehicle Owner’s Manual
2. Activate the child safety lock on one side of the vehicle at a time; the locks are typically mounted on the rear passenger doors near the latch
3. Enter the rear passenger seat of the vehicle, close the door and lock it; from inside the vehicle, attempt to open the door (it should not open with the child-safety locks activated)
4. Slide to the other side of the vehicle to exit; open the first door tested and deactivate the child-safety lock. Then repeat the procedure for the opposing door
5. Deactivate both child-safety locks when the inspection is complete

Conditions Requiring Repair:

- Door opens from the inside when the child-safety lock is activated
- Child safety lock activator is damaged or inoperative

Step 80: Window Controls

Why?

- To ensure that all manual and power window controls operate properly

How?

1. Lower and raise all door windows
2. Inspect the smoothness of operation of each window
3. Ensure that all power window switches are functional
4. Verify that the “express down” feature on power window equipped driver’s door functions properly. Depress the switch a second time to ensure that the window-lowering function stops
5. Ensure that the power window lockout feature is operational, checking the front passenger and rear window switches (they should not operate with the lockout feature activated)

Conditions Requiring Repair:

- Slow power window operation
- Malfunctioning/inoperative regulators or power window motors
- Aftermarket window tinting must be legal in the state in which the vehicle is to be sold, and must be in good condition; remove peeling, torn, bubbled, chipped, or other unacceptable tinting from the vehicle if it is to be certified

Part Four: Vehicle Interior – Door Handles, Entry, Locks, and Windows

Step 81: Remote Decklid and Fuel-Filler Door Releases

Why?

- To ensure that the manual or power remote decklid release feature operates properly on equipped vehicles
- To ensure that the remote fuel-filler door release feature operates properly on equipped vehicles

How?

1. Activate the remote decklid release lever or button and note the smoothness of operation
2. Activate the remote fuel-filler door release and note the smoothness of operation (if you are unsure of its location, consult the vehicle Owner's Manual)
3. Verify that the vehicle is equipped with an original equipment or genuine Motorcraft® gas cap

Conditions Requiring Repair:

- Decklid does not open or does not open easily when the release lever or button is activated
- Fuel-filler door does not open or does not open easily when released
- Failure of the remote decklid release or remote fuel-filler door to operate may be caused by corrosion in the release cable mechanism or the latch; consult the Service Manual for repair procedures

Note: Aftermarket gas caps must be replaced with a genuine Motorcraft® gas filler cap to be certified, because aftermarket gas caps may cause the MIL to illuminate on OBD II vehicles.

Part Four: Vehicle Interior – Luggage Compartment

Step 82: Carpet, Trim, and Cargo Net

Why?

- To ensure that luggage compartment/cargo area carpet, trim, and cargo net are in place on equipped vehicles
- To ensure that luggage compartment/cargo area carpet, trim, and cargo net are clean, undamaged, and fit properly

How?

1. Visually inspect the cargo area carpet for dirt, heavy staining, cuts, tears, or excessive wear, or for rust underneath (as possible evidence of flood damage)
2. Ensure that all trim/trim panels are properly positioned, fit properly, and are undamaged
3. Ensure that the cargo net on equipped vehicles is in place and is undamaged

Conditions Requiring Repair:

- Note any missing luggage compartment carpet or carpet that is heavily stained, heavily soiled, excessively worn, torn, or cut; replace any carpet that cannot be readily refurbished
- Missing, cracked, or broken trim or trim panels or panels that cannot be readily refurbished due to excessive soiling
- Missing or damaged cargo net

Note: Flood-damaged vehicles are not eligible for certification.

Step 83: Luggage Compartment/Cargo Area Light

Why?

- To ensure that luggage compartment/cargo area light(s) are operational

How?

1. Open the decklid, liftgate, or cargo door to confirm cargo light illumination
2. If the light is operated by an instrument panel switch or doorjamb switch, test each to confirm cargo light illumination

Conditions Requiring Repair:

- If cargo light is not operational, replace the bulb

Part Four: Vehicle Interior – Luggage Compartment

Step 84: Vehicle Jack and Tool Kit

Why?

- To ensure that the vehicle jack is in place and operates properly
- To ensure that all spare tire tools are present and in good working condition

How?

1. Locate the vehicle jack and spare tire tool kit; make sure that all tools required to change a tire are present (see the vehicle Owner's Manual for equipment location and content)
2. Remove the jack from the vehicle and ensure that it operates correctly

Conditions Requiring Repair:

- Make certain that all components are included in the vehicle
- Make certain that all components work properly in order to change a tire

Step 85: Spare Tire Size/Type and Sidewall Inspection

Why?

- To ensure that the spare tire is in place and in good condition
- To ensure that spare tire cargo area anchoring hardware is in place and operational
- To check spare tire for correct size/type and sidewall damage

How?

1. Locate the spare tire and check to see that it is properly anchored in the correct location in the vehicle and that all anchoring hardware is present/operational
2. Check the spare tire size and type against the tire size/type specified in the vehicle Owner's Manual
3. Check the spare tire tread and sidewalls for damage such as cuts, tears, heavy abrasions, rot or manufacturing defects
4. Be certain that the spare wheel rim is in good condition (not bent or heavily rusted)

Conditions Requiring Repair:

- Missing or damaged spare tire assembly or tire sidewall
- Spare tire is incorrect size or type for the vehicle
- Missing spare tire anchoring hardware
- Bent or heavily rusted spare wheel rim

Step 86: Spare Tire Tread Depth/Air Pressure Inspection

Why?

To ensure that the spare tire is in place and in good condition; full-size spare tire assemblies that do not meet minimum tread depth standards of 5/32 inch must be replaced per program policy

How?

1. For full-size spare tires, inspect the spare tire tread depth to ensure it is 5/32 inch
2. Check the spare tire air pressure and adjust it to the proper inflation level

Conditions Requiring Repair:

- Tread depth dimensions do not meet specifications for the vehicle
- Improper air pressure

Step 87: Tire Inflator Kit

Why?

- To ensure that the tire inflator kit is in place and operates properly

How?

1. Locate the tire inflator kit; make sure that all components of the kit are present. Consult the Owner's Manual for equipment location and content
2. Remove the kit from the vehicle and ensure that it operates properly

Conditions Requiring Repair:

- Make certain that all components are included in the vehicle
- Make certain that all components work properly in order to inflate a tire

Part Four: Vehicle Interior – Luggage Compartment

Step 88: Emergency Trunk Lid Release

Why?

- To ensure that the mechanical emergency trunk lid release feature operates properly on equipped vehicles

How?

1. Inspect the emergency trunk lid release for damage, including the handle, cable, and trunk lid latch
2. Fold the rear seats to access the trunk; verify proper operation of the trunk lid release by pulling on the emergency release handle

Conditions Requiring Repair:

Trunk lid does not open when the release handle is activated

Part Five: Vehicle Diagnostics Module System Test

Step 89: Perform Self-Test for All CMDTCs

Why?

- To verify that there are no stored continuous memory diagnostic trouble codes (CMDTCs) in hardware monitored by the powertrain control module (PCM)
- To ensure that any codes present are resolved prior to vehicle certification

How?

1. Connect the scan tool to the data link connector (DLC) for communication with the vehicle
2. The self-test (quick test) for all CMDTCs provides a quick check of the powertrain control system and is divided into three specialized tests:
   - Key On Engine Off (KOEO) On-Demand Self-Test
   - Key On Engine Running (KOER) On-Demand Self-Test
   - Continuous Memory Self-Test

Conditions Requiring Repair:

- Any DTCs output on the scan tool. Refer to the appropriate service materials for further diagnosis information

Part Six: Underhood – Fluids

Step 90: Engine Oil/Filter Change and Chassis Lube

Why?

- To ensure proper engine lubrication in compliance with program policy
- To ensure chassis component lubrication

How?

1. Using Motorcraft® components, perform an engine oil and filter change and inspect the used oil for fuel dilution or coolant contamination; note any contamination of the engine oil and the type of contamination if present (refer also to Steps 112 and 113)
2. Lubricate all chassis points
3. Never use nondetergent motor oil
4. Never use additional oil additives, oil treatments, or engine treatments
5. Use the proper Motorcraft® engine oil weight for safe engine operation per Ford Motor Company specifications

Step 91: Coolant

Why?

- To ensure that the coolant provides cooling system protection to the proper freeze point for your climate
- To ensure that cooling system components are in good condition
- To verify proper engine coolant type and level in the radiator and coolant recovery system

How?

1. Inspect the radiator for leaks and/or damage to the core fins
2. Ensure that the fan shroud is securely in place
3. Inspect the cooling fan for broken or bent blades
4. Verify that the proper coolant type is used; if additional coolant is required, follow the coolant refill instructions in the Service Manual
5. Use a hydrometer- or spectrometer-type coolant tester to verify that freeze protection is adequate for your climate

Conditions Requiring Repair:

- Any component leaking engine coolant
- Damaged or loose components
- Coolant that does not meet freeze protection specifications

Note: Electric cooling fans can run at any time, even with the ignition key off. Disconnect the electrical connector to the fan motor or use extreme caution when working near the fan.

Step 92: Brake Fluid

Why?

- To verify the proper brake fluid level
- To inspect for contaminated brake fluid

How?

1. Locate the brake system master cylinder and observe the fluid level through the composite reservoir; fluid should be at or near the "MAX" or "FULL" mark
2. If fluid level is low, carefully wipe the master cylinder reservoir cap clean, remove it, and fill the reservoir to the "MAX" line with genuine Motorcraft DOT 3 brake fluid
3. Be aware that an excessively low brake fluid level may indicate a leak in the brake hydraulic system; inspect the entire hydraulic system for leaks during the underbody inspection (Steps 140–169)

Conditions Requiring Repair:

- Excessively low brake fluid level in the master cylinder reservoir
- Dirty or oily-appearing brake fluid in the master cylinder reservoir may indicate that the brake hydraulic system has been contaminated by moisture or by the addition of an improper fluid such as motor oil; in these instances, flush the brake hydraulic system as directed in the Service Manual

Note: Brake fluid is toxic and removes most types of automotive paint (as well as most shop floor paint). If brake fluid is spilled on a painted surface, flush the area with water immediately and wipe it with a clean cloth.

Part Six: Underhood – Fluids

Step 93: Automatic Transaxle/Transmission Fluid

Why?

To verify proper fluid level and clarity

How?

1. Refer to workshop manual procedures for instructions on how to check for the correct level or condition of the fluid

Conditions Requiring Repair:

- Low fluid levels
- Discoloration of the fluid
- Fluid that has a burnt odor

Step 94: Transfer Case Fluid

Why?

- To verify proper fluid level and clarity
- To see whether a fluid fill or change is needed

How?

1. From underneath the vehicle, inspect the transfer case for housing cracks or any leaks; note the position and color of the fluid if leaks are present
2. Check the transfer case fluid level and color; change or fill as required. The fluid level must be just below the fill plug hole. Refer to the appropriate Service materials for transfer case fluid and filling specifications

Conditions Requiring Repair:

- Low fluid levels
- Discoloration of the fluid
- Refer to the appropriate service materials for further diagnosis to determine the extent of repairs if fluid is discolored or leaking

Step 95: Drive Axle Fluid

Why?

- To verify proper fluid level and condition of the drive axle

How?

1. From underneath the vehicle, inspect the drive axle housing for cracks or leaks; note the position and color of fluid if leaks are present
2. From underneath the vehicle, inspect the differential housing cover and seals for drips or leaks; note the position and color of fluid if leaks are present
3. Inspect the axle housing vent. A plugged axle housing vent can cause excessive pinion seal lip wear due to internal pressure buildup
4. Inspect the drive pinion seal for damage or wear

Conditions Requiring Repair:

- Refer to the appropriate Service materials for further diagnosis to determine the extent of repairs if fluid is discolored or leaking

Step 96: Power Steering Fluid

Why?

- To verify proper fluid level and clarity

How?

1. The power steering fluid level may be checked with the engine cold and/or when it is running at normal operating temperature. See specific vehicle power steering fluid filling instructions located in the vehicle Owner's Manual or in the Service Manual
2. Fill the reservoir to the appropriate level with Motorcraft® power steering fluid. Do not use ATF

Conditions Requiring Repair:

- Low fluid level (if the power steering level is low, do not drive the vehicle)
- When fluid level is low, check system components for signs of power steering fluid leakage and repair/replace components as necessary

Note: Do not turn the steering wheel with the engine off. It could force the power steering fluid out from under the reservoir cap or, in extreme cases, unseat the cap.

Part Six: Underhood – Fluids

Step 97: Manual Transaxle/Transmission Hydraulic Clutch Fluid

Why?

- To verify proper clutch fluid level on vehicles equipped with a hydraulic clutch actuator

How?

1. With the vehicle on a level surface, remove the clutch fluid reservoir cap and check to see whether the fluid is at the "MAX" level in the reservoir
2. If the fluid level is low, consult the Service Manual for instructions on refilling the reservoir

Conditions Requiring Repair:

- Low fluid level

Note: Most vehicles have see-through plastic reservoirs that permit checking fluid level without removing the cap; keep in mind that the fluid is toxic and may remove the paint finish from anything it touches. If the fluid is spilled on painted surfaces, flush the area with water immediately and wipe it with a clean cloth.

Step 98: Washer Fluid

Why?

- To verify proper washer fluid level

How?

1. Inspect the washer fluid reservoir to determine whether the fluid level is low
2. Vehicles equipped with a rear wiper/washer: Check the separate washer fluid reservoir level (refer to the vehicle Owner's Manual for specific location)
3. If the fluid level is low in either reservoir, refill them with washer fluid

Conditions Requiring Repair:

- Low washer fluid level

Note: Washer fluid contains poisonous methanol. Observe all instructions/cautions on the washer fluid bottle.

Part Six: Underhood – Engine

Step 99: Fluid Leaks

Why?

- To inspect engine components for signs of fluid leaks; to record where a leak is coming from and what is leaking

How?

1. Examine the valve cover(s) for signs of oil leakage
2. Examine the sides of the engine block for signs of coolant leakage due to a leaky or blown head gasket or rusty/perforated freeze plug(s)
3. Look for areas with a concentration of dirt/grime. This can help to identify sources of fluid leakage
4. Examine the underside of the hood and hood insulation pad for signs of oil, engine coolant, or refrigerant (air conditioner) oil; pressurized systems that leak will spray fluids – any sign of oil or coolant may indicate a system leak

Conditions Requiring Repair:

- Any component that shows signs of fluid/vacuum leakage or structural instability

Part Six: Underhood – Engine

Step 100: Hoses, Lines, and Fittings

Why?

- To verify that the following hoses, lines, and/or fittings are not leaking:
  - Coolant
  - Fuel
  - Brake
  - Steering
  - Vacuum
  - Air Conditioning

How?

1. Inspect radiator hoses and heater hoses for signs of coolant leakage or deteriorating hose clamps
2. Inspect fuel lines and fittings for abrasion, fraying, or signs of leaking O-rings. If the smell of fuel is detected, look for fuel puddling on the intake manifold (Caution: Leaking fuel is an extreme fire hazard – if a strong odor of fuel is detected, do not start the engine until the source is found and corrected)
3. Inspect the master cylinder and hydraulic clutch lines and fittings for signs of leakage or damage
4. Inspect the power steering hoses and fittings for signs of fluid leakage or deterioration
5. Inspect all vacuum lines (especially the ends) for hardening and cracking. Replace weathered or cracked vacuum hoses
6. Inspect air conditioning lines and fittings for signs of leaking refrigerant oil and failed O-rings

Conditions Requiring Repair:

Any line, hose, or fitting that is leaking must be repaired or replaced

Step 101: Belts

Why?

- To inspect engine accessory drive belt(s) for signs of cracking, fraying or accelerated wear

How?

1. Examine the serpentine belt for cracking, fraying or accelerated wear
2. Inspect the serpentine accessory drive belt tensioner to verify that it maintains its tensioning capability
3. Ensure that belts are properly adjusted

Conditions Requiring Repair:

- Any belt or drive belt component, including the serpentine tensioner, that is worn, broken or structurally unstable

Step 102: Wiring

Why?

- To verify that exposed underhood wiring is not cracked, frayed, chafed or bare

How?

1. Examine exposed underhood wiring and looms for cracking, fraying, chafing, exposed (bare), or broken wire

Conditions Requiring Repair:

- Repair or replace damaged wiring insulation or looms, and repair any bare or broken wires

Part Six: Underhood – Engine

Step 103: Oil in Air Cleaner Housing

Why?

- To check for motor oil in the air cleaner housing

How?

1. Remove the air cleaner housing lid and examine the housing interior for evidence of motor oil
2. Oil present in the air cleaner housing may indicate a possible defective PCV valve or even excessive piston ring blow-by

Conditions Requiring Repair:

- Replace a defective PCV valve
- If PCV checks OK, look for other sources of oil intrusion into the air cleaner housing. If a great deal of oil is found in the housing, a cylinder leakdown test is recommended before certifying the vehicle; ensure that the mileage indicated on the odometer is realistic for the vehicle's age. If mileage accumulation is unrealistically low (for example, 3,000 miles since the last service 18 months ago), check for signs of odometer tampering

Step 104: Water, Sludge, or Engine Coolant in Oil

Why?

- To check for evidence of water, sludge, or engine coolant in the engine oil

How?

1. Remove the oil filler cap from the engine. Inspect through the oil fill hole and inspect the underside of the cap for a milky film suggesting water, sludge, or engine coolant contamination of the engine oil
2. When the oil is being changed, inspect a sample of the used oil for evidence of water, sludge or engine coolant contamination; if oil is a milky, whitish color, consider the possibility of a blown/ leaking head gasket or water intrusion into the engine by means of a flooded roadway, etc.

Conditions Requiring Repair:

- A milky, whitish color evident on the bottom of the oil filler cap or in the used engine oil (Step 90) requires further diagnosis. Inspect engine block along bottom of the cylinder head(s) for evidence of coolant leakage, and consider performing a cylinder leakdown test on the engine before certification

Step 105: Oil Pressure

Why?

- To check for excessively low or excessively high engine oil pressure

How?

1. If the vehicle is equipped with an instrument panel-mounted oil pressure gauge, start the engine and observe oil pressure with the engine cold and at operating temperature
2. If the vehicle is equipped only with an instrument panel warning light, attach an oil pressure gauge to the oil gallery indicated in the Service Manual, checking oil pressure with the engine cold and at operating temperature
3. Oil pressure readings should be within normal ranges for the engine being tested and should increase with engine rpm until the oil pump relief pressure is reached

Conditions Requiring Repair:

- If oil pressure is excessively low, listen for connecting rod knock, ticking lifters, or other evidence that a defective oil pump, excessive bearing clearances or other major internal or external engine oil leak is present; if this is the case, major engine service may be required
- If oil pressure is excessively high, consider a sticking oil pressure bypass spring in the oil pump or restricted oil gallery passage near the gauge pickup point and look for areas of external leakage on the engine at gasket surface mating points
- Excessively low or excessively high oil pressure readings indicate that the condition must be corrected before vehicle certification

Part Six: Underhood – Engine

Step 106: Relative Cylinder Compression Test/Power Balance Readings (Check if Necessary)

Why?

- To check for evidence of compression loss due to leaking cylinder head valves, piston rings, or cylinder head gasket(s) when engine idle, performance, or driveability characteristics suggest a test is necessary

How?

1. Using the IDS, perform a relative cylinder compression test to determine the engine's general power balance condition
2. Note any areas of concern and consider performing a cylinder leakdown test if the relative cylinder compression test indicates an abnormal condition; consult the Service Manual to diagnose the condition of a weak cylinder(s)

Conditions Requiring Repair:

- Cylinder-to-cylinder variations are more likely to indicate a problem than specific pressure readings
- Note specific areas of concern for further diagnosis

Step 107: Timing Belt

Why?

- To check the condition of the timing belt
- To verify that the timing belt has been changed as directed in the vehicle maintenance schedule

How?

1. Consult the Service Manual for the specific vehicle/engine combination; inspect the timing belt as directed in the Service Manual but avoid full disassembly or removal of the timing belt cover
2. If the engine is equipped with a composite (plastic) timing belt cover, it may be possible to remove one or two retaining bolts and push the cover slightly aside to get a view of the belt. Also consider gaining access to the belt from behind the camshaft drive pulley(s) if possible. In any case, do not crack or permanently bend the timing belt cover
3. The timing belt should not be loose, frayed, cracked, or coated with grease or oil. If the belt is in good condition and is not due for replacement as directed in the vehicle maintenance schedule, move to the next step
4. If the belt is due for replacement as directed in the vehicle maintenance schedule, check to see whether the belt has been replaced. If it is clear that it has not, replace it

Conditions Requiring Repair:

- Cylinder-to-cylinder variations are more likely to indicate a problem than specific pressure readings.

Note: specific areas of concern for further diagnosis

Step 108: Engine Mounts

Why?

- To check the engine mounts for cracking, separation, breakage or leakage (hydraulic mounts)

How?

1. Visually inspect the engine mounts for evidence of cracked or torn rubber
2. On vehicles equipped with hydraulic mounts, check the surrounding area for dampness that would indicate a leak
3. On vehicles equipped with a mechanical engine cooling fan, inspect the fan shroud for cracks or missing pieces (composite shrouds) or dents (metal shrouds); inspect the fan blade edges for evidence of contact with the shroud; inspect the inboard radiator core for obvious fan damage

Conditions Requiring Repair:

- Cracked, separated, broken, or leaking engine mounts

Part Six: Underhood – Engine

Step 109: Inspect Turbocharger Air Cooler (If Equipped)

Why?

- To ensure proper operation of the turbocharger cooling system

How?

1. Inspect for leaks or weeps at the coolant expansion tank, radiator, coolant pump, fuel cooler, turbocharger actuator cooler, and all hoses
2. Inspect for damage to the coolant pump
3. Inspect for any damaged wiring or connectors
4. Verify the coolant level

Conditions Requiring Repair:

- Inoperative or malfunctioning turbocharger cooling system components
- Inoperative or malfunctioning electrical wiring or connectors
- Refer to the appropriate service materials for further diagnosis information

Part Six: Underhood – Cooling System

Step 110: Radiator

Why?

- To check for leaks and to inspect the condition of the radiator tubes and fins

How?

1. Inspect for large sections of missing, crushed, or heavily corroded fin sections
2. Inspect for leaking or poorly repaired radiator tubes
3. Inspect for leaking, dented, or damaged side tanks; for plastic tanks, check for poor retention to the core assembly and leaks, cracks, or holes
4. Check the radiator hose bungs for damage or leaks
5. On automatic transmission/transaxle-equipped vehicles, check the transmission cooler fittings/ hoses for leaks or damage

Conditions Requiring Repair:

- Large sections of missing, crushed, or heavily corroded fin sections
- Leaking or poorly repaired radiator tubes
- Leaking, dented or damaged side tanks
- On plastic side tanks, poor retention to the core assembly, leaks, cracks, or holes
- Damaged or leaking radiator hose bungs
- Leaking or damaged transmission cooler fittings or hoses

Part Six: Underhood – Cooling System

Step 111: Pressure Test Radiator Cap and Radiator

Why?

- To test the radiator cap and cooling system for leaks

How?

1. Using the proper adapter, attach the radiator cap to the pump and pressurize to test for proper operation and relief pressure
2. Remove the radiator cap and attach the pressure pump to the radiator neck
3. Pressurize the cooling system for the time specified in the vehicle Service Manual, watching for pressure drop on the gauge and leaks at the radiator core, tanks, seams, bungs, hoses, heater hoses, and heater core

Conditions Requiring Repair:

- Radiator cap that does not seal or release pressure at the pressure listed on the cap
- Any leaking cooling system component, including hose clamps

Note: Do not remove the radiator cap from a hot engine. The pressurized system raises the boiling point of the coolant, and quick removal of the pressure while the engine is hot causes the coolant to boil instantaneously, which can cause severe personal injury. Be sure the engine is cool before removing the radiator cap .

Step 112: Cooling Fans, Clutches, and Motors

Why?

- To verify that mechanical and/or electric engine cooling components operate properly

How?

1. Inspect cooling fan blades, fan clutch (if equipped), and electric fan motor (if equipped) for physical damage or looseness
2. On vehicles equipped with a mechanical cooling fan and viscous fan clutch, look for evidence of excessive leakage on the clutch, which can be a thick accumulation of grime as well as a wet look. Note that some minor leakage of silicone fluid is normal and does not mean that the clutch needs to be replaced
3. On vehicles equipped with a mechanical cooling fan and a mechanical spring-type clutch, inspect the spring for damage
4. To test fan clutch, spin the fan by hand with the engine off. If the fan will not rotate, or rotates with difficulty and a grinding noise, replace the fan clutch. If the fan freewheels (over five revolutions when spinning the fan by hand), replace the clutch
5. Electric engine cooling fans: Start the engine and turn on the air conditioning to its highest setting. This usually signals the engine management computer to request the electric fan(s) to engage. Observe fan operation and listen for any unusual whining or grinding noises. Caution: Electric fans may activate at any time – even with the engine off

Conditions Requiring Repair:

- Any damage to fan blades
- Defective fan clutch
- Defective electric fan motor

Note: Electric fans can activate at any time, even with the engine off. Keep fingers, hair, neckties, and scarves clear of all fans (mechanical or electric) or severe personal injury can result. Fan blade edges can be very sharp; wear gloves when handling them.

Step 113: Water Pump

Why?

- To check for leaks or unusual noise from the engine water pump

How?

1. With the engine running, observe operation of the water pump, checking for coolant leaks
2. Listen for unusual noise from the water pump that would suggest a defective impeller or bearing assembly, such as loud whirring, whining, or grinding noises
3. Turn off the engine. On vehicles equipped with a mechanical engine cooling fan, grasp the engine cooling fan and rock it to the front and rear of the vehicle to detect excessive bearing free play

Conditions Requiring Repair:

- Noisy, leaking, or loose water pump

Part Six: Underhood – Cooling System

Step 114: Coolant Recovery Tank

Why?

- To check for leaks
- To verify that the coolant level sensor on equipped vehicles works properly

How?

1. Visually inspect the coolant recovery tank for any physical damage that may be causing a leak
2. Inspect the area surrounding the recovery tank for evidence of a coolant leak
3. Ensure that the coolant in the tank is at the proper level for the engine temperature
4. On equipped vehicles, check the instrument panel to see whether the “low coolant” warning is displayed

Conditions Requiring Repair:

- Leaking coolant recovery tank
- Defective coolant level sensor

Step 115: Cabin Air Filter

Why?

- To verify that the cabin air filter has been changed as directed in the vehicle maintenance schedule

How?

1. Check vehicle age and mileage against the recommended cabin air filter maintenance schedule
2. If the cabin air filter is due for replacement as directed in the vehicle maintenance schedule, check to see whether the filter has been replaced. If it is clear that it has not, replace it using the appropriate Motorcraft® part as required by program policy
3. To locate the cabin air filter, consult the vehicle Owner’s Manual or Service Manual

Conditions Requiring Repair:

- Cabin air filter not replaced at the proper maintenance schedule interval

Part Six: Underhood – Fuel System

Step 116: Fuel Pump Noise Normal

Why?

- To check the fuel pump for abnormal noise
- To ensure that externally mounted pumps (if equipped) are not leaking

How?

1. Start the engine and listen for any abnormal electric fuel pump noise, such as straining or screeching sounds not consistent with normal operation (some relatively steady noise is normal)
2. Vehicles equipped with externally mounted pumps: Check for leaks around the fittings and any housing seams

Conditions Requiring Repair:

- Abnormal or excessively loud electric fuel pump noise
- Leaking externally mounted pumps/fittings (if equipped)

Step 117: Fuel Pump Pressure

Why?

- To check for correct fuel pump pressure

How?

1. Consult the vehicle Service Manual and connect a fuel pump pressure gauge to the fuel port as directed. Monitor the fuel pressure and compare against the standard listed in the Service Manual

Conditions Requiring Repair:

- Abnormally high or low fuel pump pressures
- Defective fuel pressure regulator
- Clogged/restricted fuel supply or return lines

Part Six: Underhood – Fuel System

Step 118: Fuel Filter

Why?

- To verify that the fuel filter is replaced at the proper service interval
- To replace a dirty fuel filter

How?

1. To locate the fuel filter, consult the Service Manual
2. Replace the fuel filter if it is dirty or in service beyond the proper maintenance schedule interval (use of Ford/Motorcraft® parts is required by program policy)

Conditions Requiring Repair:

- Fuel filter not replaced at the proper maintenance schedule interval
- Fuel filter is dirty in any case

Step 119: Engine Air Filter

Why?

- To verify that the engine air filter is replaced at the proper service interval
- To replace a dirty engine air filter

How?

1. Replace the air filter if it is very dirty, clogged, or in service beyond the proper maintenance schedule interval (use of Ford/Motorcraft parts is required by program policy)

Conditions Requiring Repair:

- Air filter not replaced at the proper maintenance schedule interval
- Air filter is very dirty or clogged

Part Six: Underhood – Electrical System

Step 120: Starter Operation

Why?

- To verify that the starter motor operates properly
- To check starter amperage draw

How?

1. Referring to the vehicle Service Manual or diagnostic equipment manufacturer's instruction manual, connect the vehicle starter motor to the shop's diagnostic equipment and measure starter motor draw
2. If the amperage draw is excessive, diagnose and repair or replace the starter motor as required

Conditions Requiring Repair:

- Excessive amperage draw

Step 121: Ignition System

Why?

- To verify that the ignition system operates properly

How?

1. Referring to the vehicle Service Manual or diagnostic scope manufacturer's instruction manual, connect the vehicle ignition system to the shop's diagnostic scope, and evaluate ignition system performance

Conditions Requiring Repair:

- Repair or replace any ignition component that is not operating properly as indicated by the diagnostic scope

Part Six: Underhood – Fuel System

Step 122: Battery

Why?

- To verify that the battery is correct, complete, and in good operating condition

How?

1. Verify that the battery capacity and type are correct for the vehicle in which it is installed
2. Check the condition of the terminals and the battery case. If the battery terminals themselves (not the cables) are loose/damaged, or the case/top is cracked or damaged, replace the battery
3. Look for missing vent caps and loose or missing tie-down clamps
4. Clean the battery top, cables, and posts of any corrosion, using a baking soda and water solution or commercial battery cleaner solution to neutralize any acidic corrosion
5. Check the electrolyte (fluid) level in the battery if it is not a sealed type. Note that "maintenance-free" batteries with removable cell caps may still require the addition of distilled water to the fill line; be suspicious of a nearly-dry battery and check the charging system (Step 123) before replacing the battery in this instance
6. If electrolyte is leaking, check for any wiring and/or vacuum line damage from the acid
7. Use the battery analyzer supplied by FCSD to measure the state of the battery charge. Measure the battery terminal voltage with a digital voltmeter. A battery with an open circuit voltage of 12 .6 volts is fully charged
8. Load-test the battery

Conditions Requiring Repair:

- Cracked, leaking, or missing components mean the battery must be replaced
- Low voltage will require the battery to be charged. An additional battery load test should be conducted to verify that no further battery defects exist
- **Note:** Do not use a "battery eye" (if equipped) as an indicator of state of charge.
- Be sure to wear safety glasses and protective clothing when working around a vehicle battery. If battery replacement is required, use of the proper Motorcraft® battery is required by program policy.

Step 123: Alternator Output

Why?

- To verify that the alternator output is correct

How?

1. Referring to the vehicle Service Manual for specific procedures, use battery analyzer supplied by FCSD to check alternator voltage/amperage output and verify that it is correct
2. If battery electrolyte was very low or dry in some cells, take extra care to ensure that the alternator is not overcharging the battery

Conditions Requiring Repair:

Alternator overcharging, undercharging or not charging

Step 124: Diesel Glow Plug System (If Equipped)

Why?

- To ensure that the diesel glow plug system is operating properly

How?

1. Refer to online workshop manual for glow plug system testing and operating conditions
2. Connect the scan tool to the data link connector (DLC) for communication with the vehicle
3. The glow plug monitor system is designed to detect glow plug or wiring concerns in the glow plug system. Diagnostic trouble codes (DTCs) indicate which cylinder has failed glow plugs or failed glow plug wiring

Conditions Requiring Repair:

- Inoperative or malfunctioning glow plugs. Refer to the appropriate service materials for repair information

Part Seven: Hybrid/Electric Vehicles

Step 125: Hybrid Cooling System

Why?

- To ensure the coolant is at the proper level and the proper strength for protection of the hybrid components

How?

1. The inspection and testing must be performed on a cold engine. Never open a cooling system on a hot engine
2. Visually inspect the coolant for proper level and for contamination. Use a hydrometer to ensure the coolant is at a 50/50 mixture
3. The hybrid cooling system uses the same coolant as the engine cooling system. Refer to the appropriate service materials or Owner's Manual for the coolant specifications

Conditions Requiring Repair:

- Low or contaminated coolant, and/or the coolant ratio is not at the 50/50 mixture

Step 126: Switchable Powertrain Mount

Why?

- To ensure the engine vibrations are insulated from the chassis

How?

1. Test-drive vehicle and note any abnormal vibrations
2. If there are abnormal vibrations, visually inspect the engine mount for fluid leaks or damage
3. With the engine idling, check for vacuum at the engine mount

Conditions Requiring Repair:

- Leaking or damaged mount. Refer to the appropriate service materials for repair information

Step 127: Hybrid Entertainment and Information Display

Why?

- To ensure the system is operating properly

How?

1. Bring up the energy flow information on the entertainment system
2. Test-drive the vehicle and observe the changes in power flow from the engine to the battery
3. If not operating properly, refer to the appropriate service materials for further diagnosis

Conditions Requiring Repair:

- The graphic does not switch from engine to battery when accelerating from a stop at normal operating temperature
- No information is being displayed

Step 128: Cargo Management System (BEV/Electric Focus only)

Why?

- To ensure a flat load floor in the trunk

How?

1. Check to ensure the Cargo Management System was traded in with the vehicle
2. Use the slide handle to see if the Cargo Management System is in working order with a stable flat load floor
3. Ensure the Cargo Management System does not have abnormal wear and tear

Conditions Requiring Repair:

- If the Cargo Management System is not in the vehicle or shows signs of abnormal wear, replace the missing Cargo Management System
- If the flat load floor operation does not work as intended have the system repaired

Part Seven: Hybrid/Electric Vehicles

Step 129: High-Voltage Thermal System (BEV/Electric Focus only)

Why?

- To verify that the high-voltage battery thermal system provides the proper thermal protection (cooling and heating) to protect the high-voltage battery

Note: Must be checked before the charging components tests.

How?

1. Verify that the proper coolant type is in the vehicle
2. Verify that the fluid level in the vehicle is correct
3. Inspect all of the cooling lines and joints for cracks or evidence of leaks
4. Verify that all of the thermal system lines, pumps, and the filter are properly secured and in place

Note: The high-voltage battery thermal system can operate when the vehicle is running or when the vehicle is plugged in.

Conditions Requiring Repair:

- If the incorrect fluid is in the vehicle, refer to the appropriate service materials to eliminate the problem
- Additional coolant is required
- Leaking or damaged components or signs of previous leakage
- Loose, damaged, missing, or improperly functioning components

Step 130: High-Voltage Service Disconnect (PHEV/Energi and BEV/Electric Focus)

Why?

- To verify that the high-voltage battery service disconnects are in proper working order

How?

1. Make sure that the vehicle is off and not plugged in, remove service disconnect(s) from the high-voltage battery
2. Visually inspect service disconnect for damage, bent pins, etc.
3. Reinstall service disconnect, key-up vehicle, and check BECM for any DTCs; address as needed
4. Using a diagnostic tool, collect DTCs
5. Repeat number 4 above to verify that the trouble codes have been eliminated

Conditions Requiring Repair:

- High-voltage interlock fault due to bad service disconnect, corrosion, broken lever, or bent pin
- DTCs are found

Part Seven: Hybrid/Electric Vehicles

Step 131: Charging Components – Test 1 (PHEV/Energi and BEV/Electric Focus)

Why?

- To verify that the vehicle charges properly and the consumer can view the approximate state of charge as well as whether or not the vehicle is charging

Note: This step combines the checks for these components to reduce repetitive steps if each component were checked individually.

How?

1. Ensure the 120V convenience charge cord is in the vehicle and visually inspect the cord for the integrity of the cord, the plug-in receptacle, and the cord carrying case
2. Ensure that the charge port door opens smoothly
3. Plug in the 120V charger cord and verify that there is a green light illuminated on the cord carrying case. If the light does not illuminate, do the following: Try the cord set on another vehicle; also try charging from another receptacle
4. Verify that the CPLR light sequences upper right, lower right, lower left, upper left. This sequence repeats a second time right after the vehicle has been plugged in. After this sequence the quadrant representing the approximate state of charge should continue to pulse
5. In the event the CPLR did not function as described in number 4 above, check the MyFord Touch® system (when equipped) to ensure the charge port light ring is in the "ON" setting
6. Unplug the vehicle. Place the vehicle into Accessory mode
7. Discharge the high-voltage battery to a high-voltage battery SOC that is less than 90% by running the air conditioner or heater for a period of time. If there are less than four quadrants of the CPLR illuminated in number 5 above, you can skip this step and you will be sure that you have met the criteria
8. Turn the vehicle off
9. After the vehicle is unplugged, push on the finger point on the door again, and the door should rotate back to the left and close
10. Ensure that the door remains in the closed position
11. Next, open the vehicle doors and verify that the CPLR illuminates
12. Close the doors and lock the vehicle
13. Then use the key fob to unlock the vehicle and verify that the CPLR illuminates
14. Place the vehicle in Accessory mode
15. Using a diagnostic tool, collect diagnostic trouble codes, and repair
16. Repeat number 15 above until trouble codes have been eliminated
17. Turn the vehicle off

Note: A 120V charge cord or a 240V charge cord can be used to charge the vehicle. The 120V charge cord must be used for this system check to verify that there are no issues with the cord.

Conditions Requiring Repair:

- Coolant on BEV/Focus Electric is not at an appropriate level or is contaminated
- 120V cord does not charge the vehicle
- Charge port door does not open properly
- Charge port light ring does not light or function properly
- Diagnostic codes appear

Step 132: Charging Components – Test 2 (BEV/Electric Focus only)

Why?

- To ensure that the high-voltage battery thermal system is able to moderate the temperature of the battery

Note: The 240V charge cord must be used for this system check to verify that there are no issues with the thermal system at the faster rate of charging

How?

1. Visually inspect the 240V cord for the integrity of the cord and the plug-in receptacle. Make sure that the charge station has power supplied to it, which in some cases requires flipping the power circuit on with a lever found on the outside of the station charging box
2. Discharge the high-voltage battery to a high-voltage battery state of charge that is less than 90%. Do this by running the air conditioner or heater for a period of time
3. Plug in the vehicle
4. Turn off any auxiliary loads such as heat or air conditioning and monitor the high-voltage battery state of charge using the following signals:
   - BAT_CHA_DISPL, which should be increasing
   - BAT_CHARG_STAT, which should be charging
   - BAT_CHARG_CURR, which should be less than -3 amps (240 V EVSE. A 120V EVSE may show a current between 0 and -3 amps)

Note: The high-voltage battery current will be negative when the vehicle is charging and positive when the high-voltage battery is discharging. The high-voltage battery thermal system can operate when the vehicle is in Accessory mode.

Conditions Requiring Repair:

The signals do not show the proper responses

Step 133: Cell Variation Check (BEV/Electric Focus only)

Why?

- To ensure the integrity of the high-voltage battery and to make sure that the customer is seeing the proper charge information and getting the proper range for the vehicle when the vehicle is fully charged

How?

1. Plug in the vehicle and charge the vehicle completely
2. Place the vehicle in Accessory mode
3. Read and record the following BECM PIDs:
   - HEV_BAT_MIN_V
   - HEV_BAT_VAR_V
4. Using a diagnostic tool, collect diagnostic trouble codes
5. Turn the vehicle off
6. When the trouble codes are eliminated, the sequence will need to be repeated with verification that the trouble codes have been eliminated
7. In the event that the minimum cell voltage is not greater than 4.04V and the cell variation is not less than 20mV, let the vehicle sit for 24 hours and run the test again
8. Repeat step 7

Conditions Requiring Repair:

- Diagnostic codes appear
- Minimum cell voltage is not greater than 4.04V
- Cell variation is not less than 20mV

Part Seven: Hybrid/Electric Vehicles

Step 134: DCDC (PHEV/Energi and BEV/Electric Focus)

Why?

- Ensures the DCDC is adequately charging the 12V battery

How?

1. Make sure that the vehicle has enough of a high-voltage charge that at least one of the quadrants of the CPLR is lit
2. Use a voltmeter so that the voltage on the 12V battery can be monitored
3. Check the battery terminals and ensure they are secure
4. Check the DCDC for a melted positive (B+) terminal
5. If there is an issue found with the 12V battery or the DCDC, discontinue testing and repair or replace the components
6. Start the vehicle so that it is in Run mode
7. Turn on 12V auxiliary loads such as the radio, the headlamps and the heated seats, turn the steering wheel to one side or the other, raise and lower the windows, all while monitoring the 12V battery and making sure the battery voltage doesn't drop below 12V
8. Turn the vehicle off
9. Place the vehicle in Accessory mode
10. Using a diagnostic tool, collect diagnostic trouble codes and repair
11. Repeat step 10 until trouble codes have been eliminated
12. Turn the vehicle off

Conditions Requiring Repair:

- 12V battery issue
- Diagnostic codes appear
- DCDC has a melted positive (B+) terminal

Step 135: MyFord Mobile Reset (PHEV/Energi and BEV/Electric Focus, if equipped)

Why?

To verify that the master reset on MyFord Mobile allows new customers to have owner access to the vehicle on the MyFord Mobile website

How?

1. Place the vehicle in Accessory mode
2. From the 8-inch touchscreen, press the "Menu" button
3. Press the "Setting" button
4. Press the "System" button
5. Scroll down to Master Reset (last option on the screen) and select the "Master Reset" button
6. Follow the instructions on-screen to perform the master reset
7. Turn the vehicle off
8. Place the vehicle in Accessory mode
9. Using a diagnostic tool, collect diagnostic trouble codes, and repair
10. Repeat step 9 until trouble codes have been eliminated
11. Check to make sure the MyFord Mobile-to-vehicle relationship has been reset
12. Turn the vehicle off

Note: The master reset process will take several minutes and will only function when the vehicle is stopped. The 8-inch touchscreen will reboot and power back up to the home screen as a function of resetting the master reset. The new owner can activate a MyFord Mobile account by going to myfordmobile .com and following the registration process

Conditions Requiring Repair:

- MyFord Mobile will not reset
- MyFord Mobile won't sync between the vehicle and website

Note: This will only be discovered when the new owner takes delivery and tries to connect the vehicle to MyFord Mobile.

Part Seven: Hybrid/Electric Vehicles

Step 136: Vehicle Instrument Cluster, Lifetime Summary and Trip Resets (PHEV/Energi and BEV/Electric Focus)

Why?

- Fuel economy is often the number one purchase reason for a plug-in vehicle. This process will allow new customers to obtain a new lifetime summary and track their own trip counters.

How?

1. Place the vehicle in Accessory mode
2. Using the steering wheel left-side cluster buttons, press the arrow on the left until you see the main menu
3. Using the down arrow, go to Select Settings and press the "OK" button
4. Using the down arrow, go to Display and press the "OK" button
5. Using the down arrow, go to Lifetime Summary and press the "OK" button
6. On the Lifetime Summary screen, hold down the "OK" button until you get a message indicating that a reset is in progress
7. A second message will appear indicating the reset is complete
8. Use the left arrow and return to the main menu
9. Use the down arrow and go to Trip 1 and press the "OK" button
10. On the Trip screen, hold down the "OK" button and you will be able to watch the reset occur
11. Use the left arrow and return to the main menu
12. Use the down arrow and go to Trip 2 and press the "OK" button
13. On the Trip screen, hold down the "OK" button and you will be able to watch the reset occur
14. Left arrow to the main screen
15. Turn the vehicle off

Conditions Requiring Repair:

Lifetime summary and trips do not reset

Step 137: EV Drive Modes (PHEV/C-MAX Energi/Fusion Energi only)

Why?

- Allows the driver to operate the vehicle in the three EV modes: Auto, EV Now and EV Later

How?

1. Make sure that the vehicle has been charged to a level that will allow a short amount of Auto, EV Now, and EV Later drive time (a full charge is not required)
2. Turn the vehicle on to Ready to Drive mode
3. Ensure the fuel maintenance warning and oil maintenance warning indicators do not appear; take the necessary maintenance steps if either warning appears and then restart this test
4. Push the "EV" button until you've selected EV Now mode
5. The Energy Use screen will appear on the left-hand cluster screen
6. Verify that the vehicle stays in EV Now mode for a short drive at between 40° F and 95° F

   Note: If the ambient temperature is less than 35° F, the vehicle will need to run in EV Later mode for approximately 12 to 20 minutes (or longer if the ambient temperature is in the negative range) and then the EV Now test can be completed; this allows specific parts to warm up and the test to be completed. Also, if conducted below 50° F, these component tests will cause the high-voltage battery to deplete more rapidly. Some of the drive modes may not operate at above 95° F. When testing in high temperatures, EV Now mode may not be available as a protection measure for the high-voltage battery.
7. While driving, press the left-hand steering wheel "OK" button and verify that the Energy Use screen shows the Engine Enabled message. Accelerate and verify that the engine engages. Staying at a constant speed or decelerating verifies that the vehicle goes back into EV Now mode as power demand is reduced
8. Next use the "EV" button and put the vehicle in EV Later mode
9. Verify the change to the Energy Use screen on the left-hand cluster

Step 137: EV Drive Modes, Continued (PHEV/C-MAX Energi/Fusion Energi only)

10. Drive for a short period of time and verify that the engine comes up at higher speeds, greater than 45 mph, and allows the vehicle to modulate the electric machines with the engine at speeds lower than 40 mph
11. Push the "EV" button until you've selected EV Now mode
12. Verify the change to the Energy Use screen on the left-hand cluster
13. Verify that the vehicle stays in EV Now mode for a short drive at 68° F or above ambient temperatures
14. Next use the "EV" button and put the vehicle in Auto mode
15. Verify the change to the Energy Use screen on the left-hand cluster
16. Drive the vehicle until you have used all of the EV charge
17. Next press the "EV" button and make sure the EV Now and EV Later boxes on the left-hand cluster screen have X's through them
18. Finish the drive and shut the vehicle off
19. Place the vehicle in Accessory mode
20. Using a diagnostic tool, collect diagnostic trouble codes and repair
21. Repeat step 20 until trouble codes have been eliminated
22. Turn the vehicle off

Conditions Requiring Repair:

- If any of the three modes, EV Now, EV Later, or Auto, don't work properly

Step 138: PHEV Heating System Modes (PHEV/C-MAX Energi/Fusion Energi only)

Why?

- The PHEVs are unique in that they have two heating systems: The internal combustion system provides heat when the engine is running, and the PTC heater provides heat when the vehicle is running in Electric mode

How?

1. Make sure that all air vent passages in the front and rear are unobstructed, that their directional louvers are operational, and shutoff levers are functional
2. Make sure that all air vent passage selections are operating properly
3. Make sure that the vehicle is charged such that it will run in EV Now mode for 10 to 15 minutes
4. Run this test at an ambient temperature below 80° F

   Note: If the heat is on and the ambient temperature is high, the fan speed will need to be adjusted to high in order to complete this test.
5. Turn the vehicle on to Ready to Drive and put the vehicle in EV Now mode
6. Turn the climate control system on to heat and set the cabin set point temperature to 85° F, with the blower set to low or medium speed
7. Check the air vent passages for louver operation and shutoff ability
8. Open all of the vents and check to make sure that all airflow selections are providing forced air to the corresponding vent
9. Leave the vehicle operating for 5 to 10 minutes in EV Now mode with climate settings as described above and ensure that the heating system is blowing warm air
10. Set the climate control setting to maximum defrost, without changing the cabin set point temperature
11. Allow the vehicle to run an additional 5 minutes, ensuring the heating/defrost vents are blowing hot air
12. Put the vehicle in EV Later mode

    Note: The engine should pull up; however, if there is a delay and/or the high-voltage battery has been recharged and there is still a significant charge, it may take some time before the engine pulls up. Wait until 5 minutes after the engine has pulled up to proceed to the next step.

Step 138: PHEV Heating System Modes, Continued (PHEV/C-MAX Energi/Fusion Energi only)

13. Leave the vehicle operating for 5 to 10 minutes in EV Later mode and ensure that the heating system is blowing warm air
14. Set the climate control setting to maximum defrost, without changing the cabin set point temperature
15. Allow the vehicle to run an additional 5 minutes, ensuring the heating/defrost vents are blowing hot air
16. Put the vehicle in Auto mode
17. Turn the climate control system on to auto and cycle the temperatures between 60° F, 72° F, and 85° F as set points
18. Verify the changes in the air temperatures at each of the settings as the blended air comes through the vents
19. Verify that there is a difference in the sound of the incoming air as the blower fan speed is changed from low to high
20. Shut the vehicle off
21. Check under and inside the vehicle near the front footwells for signs of coolant leakage. The signs to watch for are the smell of coolant or moisture on the carpet in the footwells
22. Place the vehicle in Accessory mode
23. Using a diagnostic tool, collect diagnostic trouble codes, and repair
24. Repeat step 23 until trouble codes have been eliminated
25. Turn the vehicle off
26. If any of the following items were encountered during this testing, the needed repairs will be required

Conditions Requiring Repair:

- Inoperative, missing or malfunctioning heater controls or vents
- Noisy or inoperative blower fan or blower switch that does not change fan speeds
- Heater blows cold air with the vehicle in EV mode for over 10 minutes
- Heater blows cold air with the engine at operating temperature
- Temperature valve adjustment does not result in a temperature change from the vent air
- Engine coolant leaking into passenger compartment (indicates probability of a heater core leak)

Part Seven: Hybrid/Electric Vehicles

Step 139: BEV Heating System Modes (BEV/Electric Focus only)

Why?

- To verify that the PTC heating system is operating properly

How?

1. Make sure that all air vent passages in the front and rear are unobstructed, that their directional louvers are operational, and shutoff levers are functional
2. Make sure that all air vent passage selections are operating properly
3. Make sure that the vehicle is charged such that it will run for 10 to 15 minutes
4. Run this test at an ambient temperature of 70° F or less
5. Turn the vehicle on to Ready to Drive
6. Turn the climate control system on to heat and set the cabin set-point temperature to 85° F, with the blower set to low or medium speed
7. Check the air vent passages for louver operation and shutoff ability
8. Open all of the vents and check to make sure that all airflow selections are providing forced air to the corresponding vent
9. Leave the vehicle operating for 5 to 10 minutes with the climate settings as described above and ensure that the heating system is blowing warm air
10. Set the climate control setting to maximum defrost, without changing the cabin set-point temperature
11. Allow the vehicle to run an additional 5 minutes ensuring the heating/defrost vents are blowing hot air
12. Set the climate control setting to maximum defrost, without changing the cabin set-point temperature
13. Allow the vehicle to run an additional 5 minutes ensuring the heating/defrost vents are blowing hot air
14. Turn the climate control system on to auto and cycle the temperatures between 60° F, 72° F, and 85° F as set points
15. Verify the changes in the air temperatures at each of the settings through the vents
16. Verify that there is a difference in the sound of the incoming air as the blower fan speed is changed from low to high
17. Shut the vehicle off
18. Place the vehicle in Accessory mode
19. Using a diagnostic tool, collect diagnostic trouble codes and repair
20. Repeat step 19 until trouble codes have been eliminated
21. Turn the vehicle off
22. If any of the following items were encountered during this testing, the needed repairs will be required

Conditions Requiring Repair:

- Inoperative, missing or malfunctioning heater controls or vents
- Noisy or inoperative blower fan or blower switch that does not change fan speeds
- Temperature valve adjustment does not result in a temperature change from the vent air

Part Eight: Underbody – Frame

(Please Place the Vehicle on a Hoist at This Time)

Step 140: Frame Damage

Why?

- To check for existing damage
- To check for signs of past repairs
- To check for abnormal tire wear

How?

1. Prior to putting the vehicle on a hoist, park on a level surface with the front wheels straight ahead. Look carefully at the vehicle from the front and rear for signs that the primary vehicle structure is bent; this is evidenced by rear tires that can be seen when looking straight ahead at the front tires (and vice versa)
2. On the hoist, visually inspect the unibody structure or vehicle frame for existing damage or signs of past repairs
3. Inspect the tires for abnormal or accelerated wear, looking for a specific, uneven wear pattern

Conditions Requiring Repair:

- Vehicles with bent frames or unibody structures are not eligible for certification

Step 141: Fuel Supply System

Why?

- To check for leaks or damage to:
  - Fuel lines, fuel tank, fuel hoses, and couplings

How?

1. Inspect the fuel tank for damage or leaks and check the fuel hoses at the tank for deterioration or leaking; check all crimped or clamped hose-to-hardline connections for leaks
2. Check the condition of the fuel lines along the length of the vehicle, inspecting for leaks, rust-through, kinking, or loose anchor clips

Conditions Requiring Repair:

- Replace damaged fuel tanks, heavily rusted or leaking fuel lines and clamps, as well as cut, cracked, or leaking fuel hoses and couplings

Step 142: Exhaust System Condition

Why?

- To check the exhaust system for leaks or damage

How?

1. Starting at the exhaust manifold(s), inspect the exhaust pipe(s), catalytic converter(s), muffler(s), and resonator(s) (if equipped) for severe rust, road hazard damage, holes, and leaks; inspect all coupling connection points for exhaust gas leaks
2. Check air injection tubes at the catalytic converter for proper anchoring and ensure that they are not leaking
3. Look for missing, severely rusted or broken hangers and clamps

Conditions Requiring Repair:

- Replace all missing, severely rusted, broken, or leaking exhaust system components

Part Eight: Underbody – Exhaust System

Step 143: Emissions Control Test/Emission System Inspection

Why?

- To ensure that the vehicle emission controls are operational and meet federal, state, and local government exhaust emission standards

How?

1. Inspect major emission components such as the PCV valve, EGR valve, underhood air injector pump/lines, exhaust manifold connections, exhaust pipe(s), muffler(s), catalytic converter(s), oxygen sensor(s) – both fore and aft of the catalytic converter(s) – and the OBD II control system's PCM to ensure that they are attached, operating, and show no signs of damage
2. The "Check Engine" light illuminates to indicate any emission system diagnostic codes
3. Perform an emissions test to ensure that the vehicle complies with all local, state, and federal emissions requirements as applicable. All Ford products are designed to pass these tests. If a Ford product fails an I/M test, it is probably because the engine or catalyst temperature was not warm and stabilized before the test, or the vehicle had idled excessively before the test

Conditions Requiring Repair:

- Failed emissions test. Refer to the appropriate service materials for further diagnosis information
- Damaged or deteriorating emission components. If any new emission components are installed, carry out the following steps before repeating the I/M test procedure: Reset the keep alive memory (KAM); relearn some basic adaptive learning (trim) values by running the engine at 2,500 rpm for one minute, and idling the engine for two minutes
- Following repairs, clear all codes from memory, test-drive the vehicle and, using the scan tool, recheck for codes that may have reset

Part Eight: Underbody – Transmission, Transaxle, Differential, and Transfer Case

Step 144: Automatic Transmission/Transaxle

Why?

- To check the automatic transmission/transaxle for leaks

How?

1. Inspect the transmission cooler lines and fittings at the radiator and at the transmission case for leaks
2. Inspect the torque converter dust cover vents and edges for drips and leaks
3. Inspect the transmission/transaxle shift quadrant (shift cable input) and output shaft(s) for leaks
4. Inspect the transmission fluid reservoir pan for leaks due to overtorqued bolts, mushroomed bolt holes or a failed gasket

Conditions Requiring Repair:

- Note the position of any leaks and the color of the fluid
- Further diagnosis is required to determine the extent of repairs required if fluid/lube is discolored or leaking

Step 145: Manual Transmission/Transaxle, Differential and, Transfer Case

Why?

- To check the manual transmission/transaxle, differential, and transfer case (4WD vehicles) for leaks

How?

1. From underneath the vehicle, inspect the manual transmission/transaxle, differential, and transfer case (4WD vehicles) for housing cracks or for any leaks; note their position and the color of the fluid if leaks are present

Conditions Requiring Repair:

- Note the position of any leaks and the color of the fluid/lube
- Further diagnosis is required to determine the extent of repairs required if fluid/lube is discolored or leaking

Part Eight: Underbody – Transmission, Transaxle, Differential, and Transfer Case, Continued

Step 146: 4x4 Hub Operation

Why?

- To ensure the 4x4 hubs engage and disengage properly

How?

1. Test-drive the vehicle. The operator can switch between two-wheel-drive high (2H), Neutral (N), four-wheel-drive high (4H) and four-wheel-drive low (4L) at speeds below 5 km/h (3 mph). To engage or disengage 4L, the transmission should be in Neutral to alleviate residual drag torque created by the engine
2. Prior to shifting to 4H and 4L, the front axle must be coupled to the hubs by rotating the hublocks fully clockwise to the "LOCK" position; verify the hubs are properly engaged
3. Check for mechanical engagement of 4H and 4L; drive the vehicle in tight turns on a hard surface. Verify the presence of driveline windup and tire scuff
4. Prior to shifting to 2H, rotate the hublocks fully counterclockwise to the "FREE" position; verify the hubs are properly disengaged

Conditions Requiring Repair:

- Manual hublocks do not engage in the "LOCK" position
- Manual hublocks do not disengage in the "FREE" position
- Refer to the appropriate Service materials for further diagnosis information

Step 147: Universal Joints, CV Joints, and CV Joint Boots

Why?

- To check the universal joints, CV joints and CV joint boots for proper positioning, damage, leaks or excess wear/loosenessHow?

1. Visually inspect the condition of the driveshaft universal joints or CV joints for damage or excessive wear and check for excess looseness by rocking the attached shaft in a circular pattern; if the joint checks OK and no driveline "clunk" was detected during the road test, lubricate if possible
2. Check the CV joint boots for proper positioning
3. Check the condition of CV joint boots for cuts, tearing, or excess wear; replace if necessary

Conditions Requiring Repair:

- Excessively worn or damaged universal joints, CV joints, or CV Joint Boots

Step 148: Transmission Mounts (not cracked, broken, or oil soaked)

Why?

- To check the transmission mounts for cracking, separation, breakage, or fluid

How?

1. Inspect the transmission mounts for cracking, breaks, or fluid
2. Check the surrounding area for signs of leakage

Conditions Requiring Repair:

- Cracked, broken, or fluid-soaked transmission mounts. Refer to the appropriate service materials for repair information

Part Eight: Underbody – Transmission, Transaxle, Differential, and Transfer Case

Step 149: Differential/Drive Axle

Why?

- To check the differential/drive axle for damage or leaks
- To inspect all drive axle housings for proper operation and structural soundness
- To inspect front (4WD models) and rear (RWD) differentials for leaks, proper fluid levels, and fluid quality

How?

1. Visually inspect the condition of the differential/drive axle assembly/ assemblies for road hazard damage, cracks, or leaks
2. If leaks are present, use a siphon to extract a sample of the axle lube, and examine the lube to see whether water is present
3. Make sure that the fluid level is within 1/4 inch to 9/16 inch of the bottom of the fill hole

Conditions Requiring Repair:

- If a leak is detected, identify the area of the leak. Note whether the leak is caused by a crack in the differential housing, a worn gasket, or a worn pinion seal
- Correct the leak and fill the differential to the proper level

Part Eight: Underbody – Tires And Wheels

Step 150: Tires and Wheels Match and Are Correct Size

Why?

- To verify that all tires and wheels on the vehicle are the same brand, type, and size
- To verify that all hubcaps/wheel center hubs are present
- To verify that tires are not involved in a recall

How?

1. Visually inspect the tires and wheels on the vehicle to ensure that they are all of the same type and size and that the tires and wheels are the correct size for the vehicle (consult the vehicle Owner's Manual, Ford Accessories Catalog and/or the Ford Aftermarket Wheel Policy)
2. Verify that all hubcaps and wheel center hubs are present
3. Ensure that all wheels have the correct number of lug nuts and that no wheel studs are broken or missing
4. Verify that directional tires (if equipped) are properly installed on the correct side or corner of the vehicle
5. Using OASIS, check for and complete any open tire-related recalls on the subject vehicle

Conditions Requiring Repair:

- Mixed wheel types, tire brands/sizes and/or incorrect size tires/wheels on the vehicle must be corrected before the vehicle can be certified
- All mounting hardware must be in sound condition
- Directional tires must be oriented to the proper side/corner of the vehicle

Note: When the wheels/tires are removed from the vehicle for the brake inspection (Steps 164–169), consider rotating them when reinstalling. Check the Service Manual or vehicle Owner's Manual for recommended tire rotation direction.

Part Eight: Underbody – Tires and Wheels

Step 151: Tire Tread Depth

Why?

- To verify that tire tread depth meets program standards

How?

1. Using a machinist's rule or the tire tread depth gauge provided by Ford Customer Service Division, measure the tread depth on all tires, including the spare
2. Record the tread depth of each tire and note on the Inspection Checklist

Conditions Requiring Repair:

- Replace a tire with the correct tire if tread depth is less than 5/32 inch

Step 152: Normal Tire Wear

Why?

- To verify that tire wear pattern(s) are normal, with no alignment problems or sidewall defects

How?

1. Inspect for unusual or uneven tire tread wear patterns
2. Check the tire sidewalls for bulges, cuts, tears, or tread separation and replace the tire if any of these conditions are present

Conditions Requiring Repair:

- Vehicles with tires displaying an unusual wear pattern should have the wheel alignment inspected
- Replace any tire with bulges, cuts, tears, or tread separation
- Ensure that the tire wear bars are not visible, regardless of measured tread depth

Step 153: Tire Pressure

Why?

- To verify that tire air pressures are correct

How?

1. Using an accurate tire pressure gauge, measure, and note the air pressure in each tire
2. Refer to the vehicle label identifying proper tire pressure and check OASIS to verify that tire pressure recommendations have not been revised for the vehicle; adjust the air pressure to the correct reading
3. Ensure that all tire pressures are the same and check tires when cold

Conditions Requiring Repair:

- Incorrect tire pressure

Part Eight: Underbody – Tires and Wheels

Step 154: Tire Pressure Monitoring System

Why?

- To check that the Tire Pressure Monitoring System (TPMS) monitors the air pressure of all four road tires and alerts the driver to low tire pressure conditions by illuminating the TPMS indicator

How?

1. Tire pressures fluctuate with temperature changes. For this reason, tire pressure must be set to specification when tires are at outdoor ambient temperatures
2. If the spare tire is in use, the damaged road tire must be repaired and installed on the vehicle to restore complete TPMS functionality before carrying out any diagnosis
3. Verify that the correct OEM wheels and tires are installed
4. For vehicles with different front- and rear-tire pressures (such as E-Series and certain F-Series), the tire pressure sensors must be trained following a tire rotation. Failure to train the sensors will result in a false low tire pressure event, which will cause the TPMS indicator to illuminate
5. Using an accurate tire pressure gauge, measure, and note the air pressure in each tire. Verify the proper air pressures by consulting the tire and information label or the Owner's Manual. Compare air pressures to message center readings (if equipped)
6. Inspect the tire pressure sensors for correct application and for damage
7. If the TPMS indicator flashes for 70 seconds, and then remains on when the ignition is turned to the "ON" position, check the appropriate service materials for diagnosis information

Conditions Requiring Repair:

- Damaged or incorrect tire pressure sensors
- No tire pressure readings or inaccurate tire pressure readings. Refer to the appropriate service materials for further diagnosis

Step 155: Wheels, Wheel Covers, and Center Caps

Why?

- To check the condition of the valve stems/caps
- To verify that the wheels and wheel finishes are free from damage
- To ensure that wheels are not bent and do not have excessive runout
- To check the condition of the wheel covers or center caps

How?

1. Check the valve stems to ensure that they are in good condition and that all caps are in place; replace missing caps
2. Visually inspect the wheels and hubcaps (if equipped) for structural damage
3. On vehicles equipped with road wheels, inspect for damage to the painted or chromed finish
4. Visually inspect the rim at the tire bead around the circumference of the wheel on the inboard and outboard sides of the rim for dents, bending, or other physical damage
5. Spin each wheel/tire assembly by hand and watch both inboard and outboard sides of the rim at the tire bead for wobble or excessive runout (it is not necessary to use a dial gauge for this step, as some minor runout is normal)
6. Visually inspect the wheel covers or center caps for damage

Conditions Requiring Repair:

- Damaged valve stems and missing valve stem caps
- Missing hubcaps
- Bent wheels
- Damage to painted or chrome finish on road wheels
- Wheels with excessive runout
- Missing wheel covers or center caps
- Loose or damaged wheel covers or center caps

Part Eight: Underbody – Suspension and Steering

Step 156: Rack-and-Pinion, Linkage, and Boots

Why?

- To check the rack-and-pinion assembly, linkage, or boots for damage or leakage

How?

1. Inspect the power-assisted steering system for leaks or physical damage
2. Inspect the steering rack boots for cuts, tears, or heavy abrasions and replace as necessary
3. Inspect the steering linkage for physical damage due to impacts or road hazards

Conditions Requiring Repair:

- Leaking steering assembly
- Physical damage to the rack-and-pinion assembly
- Damaged rack-and-pinion system dust boots
- Physical damage or looseness in the steering linkage

Step 157: Control Arms, Bushings and Ball Joints

Why?

- To ensure proper positioning and mounting of key suspension components
- To check for physical damage or deterioration to control arms, bushings, and ball joints
- To check for excessive play in the ball joints

How?

1. Check suspension control arms for proper positioning and mounting
2. Inspect control arms for physical damage and, if applicable, note which control arm is damaged
3. Check control arm bushings for dry rot, tearing, or shrinkage; ensure that they are securely anchored/ pressed into position
4. Inspect ball joint grease boots for cuts, tearing, or deterioration
5. Inspect ball joints for physical damage
6. Inspect ball joints for broken, clogged or missing zerk (grease) fittings, or missing anchor rivets/bolts or poor press-fit as applicable
7. Consult the Service Manual for full procedures to inspect for ball joint assembly wear

Conditions Requiring Repair:

- Improperly positioned or mounted suspension components, including control arms, linkages, bushings, and ball joints
- Physical damage to any listed suspension component
- Cut, torn, shrunk or rotting bushings or grease boots
- Inoperative or missing zerk (grease) fittings
- Worn ball joint assemblies

Part Eight: Underbody – Suspension and Steering

Step 158: Tie Rods and Idler Arm

Why?

- To ensure that the steering system's tie rods and idler arm are in serviceable condition
- To check for physical damage or deterioration to tie-rods or idler arm

How?

1. Visually inspect the inner and outer tie-rods for proper positioning and mounting
2. Ensure that the tie-rods are not severely rusted and inspect the adjusting sleeves and clamps to ensure that they are not damaged or seized
3. Inspect the tie-rod grease boots for cuts, tears, or heavy abrasions and replace if any are present
4. Inspect the idler arm for proper positioning and mounting
5. Inspect the idler arm grease boot for cuts, tears, or heavy abrasions and replace if any are present
6. Check the idler arm for excessive wear and replace as necessary

Conditions Requiring Repair:

- Improperly positioned or mounted tie-rods or idler arm
- Tie-rod or idler arm grease boots are torn, cut, or heavily abraded
- Tie-rod threads or adjusting sleeves are damaged, severely rusted or seized, preventing wheel alignment adjustments from being made
- Worn tie-rods or an idler arm that does not operate to Service Manual specifications

Step 159: Sway Bars, Links, and Bushings

Why?

- To ensure that the vehicle's sway bars (stabilizer bars), end links, and associated bushings operate to specifications

How?

1. Visually inspect all grommets and bushings for deterioration (cracks/tears/rot) or excessive wear
2. Ensure that the front sway bar's front anchor bushings are firmly secured
3. Ensure that the front sway bar's end link hardware and bushings are not severely rusted, bent, or pulled out of their mounts
4. Inspect the front sway bar for bends or other physical/road hazard damage
5. Inspect the rear sway bar (if equipped) for proper positioning and mounting
6. Inspect the rear sway bar for bends or other physical/road hazard damage

Conditions Requiring Repair:

- Cut, cracked, torn, or rotted grommets or bushings must be replaced
- Severely rusted or damaged front sway bar end link hardware must be replaced
- Bent/damaged sway bars must be replaced

Part Eight: Underbody – Suspension and Steering

Step 160: Springs

Why?

- To ensure that the vehicle's springs are not broken or sagged and provide the proper ride height

How?

1. Visually inspect all springs for physical damage or breakage; replace in pairs as necessary
2. Inspect for any non-original equipment spring helpers, such as rubber shims/spacers, screw adjusters, helper springs, non-OE air-type or coil-over-type shock absorbers, etc. Remove them and restore the vehicle to the correct ride height using new springs
3. Ensure that the vehicle ride height is acceptable and that the springs are not sagged; replace as necessary in pairs to restore proper ride height

Conditions Requiring Repair:

- Broken or sagged springs; replace in pairs as necessary
- Non-OE spring helpers as outlined above

Step 161: Struts and Shocks

Why?

- To ensure that the shock absorbers and struts are not leaking and are in good condition

How?

1. Inspect all shock and strut assemblies for fluid loss, which indicates seal damage in the component. As fluid leaks from the unit, shock-damping ability decreases
2. Inspect for severe rust or damage that compromises the structure of the shock or strut

Conditions Requiring Repair:

- Worn, broken, bent, or leaking shocks or struts must be replaced

Step 162: Wheel Alignment (check if necessary)

Why?

- To ensure that the wheel alignment is correct

How?

1. If, during the road test, the vehicle exhibited any pulling (brakes not applied) or wandering, check and correct the wheel alignment
2. If any abnormal tire tread wear is present, check and correct the wheel alignment
3. If any steering system components required replacement during the inspection, check and correct the wheel alignment following service

Conditions Requiring Repair:

- Pulling or wandering when the brakes are not applied
- When abnormal tire tread wear is present
- When steering system components are replaced

Part Eight: Underbody – Suspension and Steering

Step 163: Power Steering Pump

Why?

- To check the power steering pump for leaks or unusual noises

How?

1. With the engine running, observe and listen to the power steering pump. Note any whining, grinding, or whirring noises coming from the pump. Because the fluid level was checked and/or corrected in Step 96, and the drive belt inspected in Step 101, any noises are likely to be coming from the unit itself and should be noted on the checklist
2. Inspect for leaks around the filler neck, pressure and return hoses/fittings and the reservoir itself. Note any drips/leaks on the checklist
3. Ensure that the power steering drive pulley tracks smoothly as it rotates. Correct/replace a loose, bent or out-of-round pulley

Conditions Requiring Repair:

- A noisy pump not due to low fluid level
- Leaking filler neck, pressure and return hoses/fittings and/or reservoir
- Loose, bent or out-of-round pump drive pulley

Part Eight: Underbody – Brakes

Step 164: Calipers and Wheel Cylinders

Why?

- To check calipers and wheel cylinders for proper operation

How?

1. Remove the tire and wheel assemblies from the vehicle
2. Inspect the brake calipers for leaks or evidence of seizing (one pair of pads worn more than the other pair); correct as necessary
3. Remove the brake drums and inspect the wheel cylinders for leaks; correct as necessary
4. Inspect the wheel cylinders for proper operation; rebuild or replace as necessary

Conditions Requiring Repair:

- Leaking or seized brake caliper pistons
- Seized brake caliper sliders or anchor pins
- Leaking or inoperative wheel cylinders
- Physically damaged brake calipers or wheel cylinders, including damaged bleeder screws

Part Eight: Underbody – Brakes

Step 165: Brake Pads and Shoes

Why?

- To determine that the brake shoes and pads meet program brake pad/lining thickness standards
- To check for unusual wear characteristics

How?

1. Clean the brake component surfaces to remove any dust or dirt
2. Ensure that all brake components are in their proper positions and move freely
3. Inspect and measure the brake pads and shoes; record measurements on the Inspection Checklist. Pads and shoes that do not meet program brake pad/lining thickness standards must be replaced per program policy
   - Brake pads/linings must have at least 50% thickness remaining

Conditions Requiring Repair:

- Brake shoes or pads that do not meet minimum thickness standards (also replace brake springs/retaining clips when shoes/pads are replaced)
- Any wheel brake with unusual wear characteristics, such as excessive runout or brake fluid contamination of the friction lining
- Seized starwheel adjusters
- Any bent or broken brake hardware, including brake springs, pins, anchors, links, and adjusters
- Physically damaged brake calipers or wheel cylinders, including damaged bleeder screws

Step 166: Rotors and Drums

Why?

- To determine that the brake rotors and drums have at least the minimum allowable thickness of their allowable wear life remaining
- To check for scoring of the rotor or drum friction surface

How?

1. Clean the brake component surfaces to remove any dust or dirt
2. Measure the thickness of rotors with a micrometer and determine whether thickness is greater than minimum thickness specification
3. Using an inside caliper, measure the inside diameter of the brake drums and determine whether thickness is greater than minimum thickness specification
4. Check rotors and drums for scoring
5. Check for runout beyond allowable Service Manual specifications

Conditions Requiring Repair:

- Brake rotors or drums with less than the minimum allowable thickness must be replaced.
- Scored, gouged, pitted or warped rotors or drums must be machined or replaced
- Correct situations where runout is beyond allowable Service Manual specifications

Part Eight: Underbody – Brakes

Step 167: Brake Lines, Hoses, and Fittings

Why?

- To determine the condition of brake lines, hoses, and fittings
- To inspect for leaks

How?

1. Inspect brake hose-to-hardline fittings for leaks, brake hose anchor clips for deterioration, and all other fittings for leaks in such locations as:
   - Proportioning valve
   - Combination valve
   - ABS modulator assembly or pressure relief reservoir (as equipped)
   - Calipers and wheel cylinders
2. Inspect rubber brake hose for cracking, hardness, or leaks
3. Inspect all brake pipes (hardlines) for severe rust, kinks, breaks, or other leak sources

Conditions Requiring Repair:

- Replace all cracked, hard, rotting, or leaking rubber brake hoses and/or weak anchor clips
- Replace brake pipes that are severely rusted, kinked, broken, or leaking
- Repair leaking fittings

Step 168: Parking Brake

Why?

- To ensure proper parking brake operation
- To determine the condition of parking brake components

How?

1. Inspect the parking brake cable for fraying and proper tension
2. Inspect the parking brake cable anchor spring clips ("fingers") to ensure that they are secure in their holes; replace corresponding portion(s) of the cable assembly as necessary
3. Replace missing or severely rusted cable anchor clips or wires as necessary
4. With the brake drums removed, inspect the parking brake's wheel components for seizing or binding and replace as necessary
5. Test for proper operation and make any necessary adjustments

Conditions Requiring Repair:

- Broken, frayed, or damaged parking brake cable
- Missing/severely rusted cable anchor spring clips ("fingers") that allow the parking brake cable to move freely within its anchor hole require replacement of the corresponding portion of the cable assembly
- Missing/severely rusted cable anchor clips/wires
- Seizing or binding parking brake wheel components
- Loose or overly tight cable adjustment

Step 169: Master Cylinder and Booster

Why?

- To determine the condition of the master cylinder and brake booster

How?

1. Inspect the master cylinder and booster for excessive corrosion or damage
2. Inspect the booster for leaks at vacuum connections
3. Inspect the vacuum hoses for leaks or kinks
4. Verify the operation of the check valve
5. The following conditions are considered normal and are not indications that the brake master cylinder is in need of service:
   - Condition 1: During normal operation of the brake master cylinder, the fluid level in the brake master cylinder reservoir will fall during brake application and rise during release.
     The net fluid level (such as after brake application and release) will remain unchanged
   - Condition 2: Fluid level will decrease with pad wear
   - Condition 3: On vehicles equipped with a vacuum booster, a trace of brake fluid will exist on the booster shell below the master cylinder mounting flange. This results from the normal lubricating action of the master cylinder bore and seal

Conditions Requiring Repair:

- Replace all cracked, damaged, or leaking components
- Leaking or damaged master cylinder booster

Part Nine: Convenience

Step 170: Owner's Manual

Why?

- Verify the presence of the Owner's Manual in the vehicle

How?

1. Check that the Owner's Manual is present in the vehicle glove box
2. Check that the Owner's Manual includes all appropriate materials

Conditions Requiring Repair:

- Replace any missing materials from the Owner's Manual

Step 171: Keys and Remote Controls

Why?

- Verify the presence of all keys and remote controls (key fobs)

How?

1. Check that all appropriate keys, including ignition keys, door keys, tailgate or accessory keys, and glove box keys are with the vehicle
2. Check that all appropriate remote controls (key fobs) are with the vehicle

Conditions Requiring Repair:

- Replace any missing keys
- Replace any missing remote controls and program to the vehicle
- Refer to the appropriate service materials for programming information

Step 172: Full Fuel Level

Why?

- To verify the fuel tank is at the full level

How?

1. Check the fuel gauge to ensure that it reads full

Conditions Requiring Repair:

- A fuel gauge reading less than full
- Fill the fuel tank with the required fuel as needed